Politecnico di Torino

Master's Degree in Mathematical Engineering
A.a. 2025/2026
Graduation Session March 2026

DNA Methylation Dynamics Along the Normal–Adjacent Axis in Breast Cancer

A Quantitative Statistical and Machine Learning Approach

Supervisors:
Prof. Ing. Alfredo BENSO
Dott. Sandro GAMBINO

Candidate:
Elisabetta ROVIERA

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3.1 Methodological Objectives

This chapter establishes the methodological foundation for all subsequent analyses by addressing the construction, organization, and preliminary characterization of the DNA methylation datasets used in this thesis. Following the biological background and problem formulation introduced in Chapters 1 and 2, the present chapter acts as a bridge between domain-specific knowledge and quantitative modeling, ensuring that downstream analyses are grounded on a technically sound and well-understood data representation.

Genome-wide DNA methylation data are characterized by extremely high dimensionality, heterogeneous signal-to-noise ratios, and the coexistence of subtle biological effects with multiple sources of technical variability. In this context, premature application of statistical models or feature-selection procedures may lead to biased conclusions driven by artefacts rather than genuine biological structure. A careful examination of data quality, distributional properties, and internal consistency is therefore a necessary prerequisite for any reliable inference [1].

The primary objective of this chapter is to construct a coherent and reproducible representation of the available datasets and to assess their structural and statistical properties prior to any form of preprocessing or modeling. Specifically, this chapter aims to: (i) describe the organization of methylation matrices and associated phenotype information; (ii) characterize the global and local variability of methylation levels within each dataset; (iii) evaluate the degree of separation between Normal, Adjacent, and Tumor tissues using exploratory tools; and (iv) compare datasets at an inter-cohort level to assess consistency, heterogeneity, and potential limitations in signal transferability.

The scope of this chapter is deliberately restricted to exploratory and diagnostic analyses. No feature selection, predictive modeling, or formal statistical inference is performed at this stage. All analyses are intended to be descriptive and comparative, serving to highlight structural properties of the data and to identify potential technical or biological issues that must be addressed before proceeding further.

Methodologically, the chapter is organized around a unified exploratory framework applied consistently across all datasets. Intra-dataset analyses are used to investigate internal structure, variability patterns, and group-level behavior, while inter-dataset comparisons focus on assessing cross-cohort consistency and platform-related differences. Throughout the chapter, visualizations and summary metrics are employed as diagnostic tools rather than as decision-making instruments.

The results of this exploratory phase reveal both shared characteristics and dataset-specific behaviors, as well as clear limitations of working directly with raw or minimally processed methylation data. These observations motivate the need for a structured and rigorous data pre-processing pipeline, which is developed and justified in the following chapter. The present chapter therefore provides the conceptual and technical basis for the preprocessing choices adopted in Chapter 4 and for all subsequent modeling steps.

3.2 Dataset Construction, Metadata Integration, and Data Organization

This section provides the methodological framework for the selection, construction of the dataset, and the organisation process adopted in this study. It contextualises the data sources, representation choices, metadata management, and computational design principles that underpin the detailed implementation described in the following sections.

3.2.1 Data Sources and Study Design

All datasets analyzed in this thesis were obtained from the NCBI Gene Expression Omnibus (GEO) [2], which was selected as a unified data source to minimize heterogeneity in data formats, metadata conventions, and access mechanisms. GEO provides curated series-level accessions, stable sample identifiers, and standardized links to processed methylation matrices and sample-level annotations, making it a natural reference repository for large-scale DNA methylation studies.

The study design adopts a multi-cohort comparative framework focused on genome-wide DNA methylation in breast tissue. Three independent GEO series were selected for analysis: **GSE69914** [3], based on the Illumina HumanMethylation450 platform, and two more recent EPIC-based cohorts, **GSE225845** [4] and

GSE287331 [5]. Together, these datasets span normal/healthy, adjacent-normal, and tumor tissue states across two generations of Illumina methylation arrays (450K and EPIC), enabling both within-dataset characterization and cross-dataset comparisons.

Dataset selection was driven by strict inclusion criteria aimed at ensuring technical compatibility, biological relevance, and analytical feasibility. In particular, only studies providing processed genome-wide methylation matrices in terms of β-values were considered, allowing a uniform data representation and avoiding the need for low-level signal reprocessing from raw `.idat` intensity files. Raw `.idat` files store probe-level fluorescence intensities measured in the methylated ($y^{(M)}$) and unmethylated ($y^{(U)}$) channels, from which β-values are derived as a normalized ratio [6]:

$$\beta := \frac{\max(y^{(M)},0)}{\max(y^{(M)},0) + \max(y^{(U)},0) + \alpha}, \qquad \beta \in [0,1]. \tag{3.1}$$

where α is a small offset (typically 100) used to stabilize the denominator. Additional requirements included the use of Illumina HumanMethylation450 (450K) or MethylationEPIC (EPIC) platforms, the availability of multiple tissue classes within each cohort, and sufficiently large sample sizes to support robust statistical analysis.

3.2.2 Methylation Data Representation and Storage

In several datasets, the original authors applied established normalization procedures, including methods specifically designed to correct probe-type design bias such as BMIQ [7]. Reprocessing raw IDAT files would therefore not add information relevant to the objectives of this thesis, while substantially increasing technical complexity and reducing reproducibility. For these reasons, all analyses were conducted on the processed β-value matrices as distributed via GEO.

To support scalable analysis and efficient input/output operations, all working methylation matrices were converted to a standardized `sample` × `CpG` layout and stored in columnar Parquet format using single-precision floating-point encoding.

This choice is motivated by both numerical and computational considerations. Since β-values are strictly bounded in the interval [0,1] [6], and biologically meaningful methylation differences typically occur at magnitudes on the order of 10^{-2}–10^{-3}, single-precision floating point (`float32`, machine $\varepsilon \approx 10^{-7}$) provides more than sufficient numerical accuracy. At the same time, it substantially reduces memory usage and I/O overhead compared to double precision.

This representation is consistent with recent large-scale genomics frameworks that process molecular features, including DNA methylation data, entirely in `float32` precision for efficiency and scalability [8].

3.2.3 Phenotype Tables and Label Harmonization

For each dataset, a dedicated phenotype table was constructed by extracting and parsing GEO sample-level metadata. These tables encode tissue labels, sample identifiers, and relevant clinical or technical covariates when available. Given the heterogeneity of original annotations across studies, tissue classes were mapped to harmonized numeric encodings to ensure consistency across cohorts while preserving dataset-specific attributes in separate fields.

A strict one-to-one correspondence between methylation matrices and phenotype tables was enforced. Samples lacking either methylation measurements or corresponding metadata were excluded to prevent inconsistencies and downstream misalignment. This design guarantees that all analyses operate on synchronized molecular and phenotypic representations and enables reliable stratification by tissue class in both intra- and inter-dataset comparisons.

3.2.4 Reproducibility and Computational Workflow

The overall data organization and computational design follow a multi-dataset analytical paradigm commonly adopted in recent large-scale DNA methylation studies, in which cohort-specific analyses are complemented by cross-dataset comparisons to explicitly evaluate robustness and generalizability, while simultaneously revealing cohort-dependent effects and limitations in signal transferability that motivate subsequent methodological refinement [9].

All dataset construction and organization steps were implemented using deterministic, script-based pipelines with explicit schema definitions and typed data representations. Particular care was taken to avoid unnecessary in-memory operations on high-dimensional methylation matrices, favoring streaming ingestion, chunked processing, and out-of-core transformations when required by dataset size.

The resulting data organization yields a set of standardized, self-contained datasets, each consisting of a genome-wide methylation matrix and an aligned phenotype table stored in efficient, portable formats. This design ensures full reproducibility of the data preparation process and provides a robust foundation for the preprocessing, exploratory analysis, and modeling steps developed in subsequent chapters.

3.3 Intra-Dataset Exploration and Visualization

This section presents a structured intra-dataset exploratory analysis aimed at characterising the internal statistical and biological organisation of each methylation cohort prior to any preprocessing or modelling step. Each dataset is analysed independently in order to assess data integrity, quantify variability patterns, and

Table 3.1: Overview of the GSE69914 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor	Normal (BRCA1)	Tumor (BRCA1)
485,512	407	50	42	305	3	7

evaluate whether Normal, Adjacent, and Tumor tissues exhibit distinguishable epigenetic behaviour.

In the context of breast cancer, epigenetic deregulation is known to manifest through both global and local phenomena, including genome-wide hypomethylation, focal hypermethylation of tumour-suppressor regions, and the presence of pre-neoplastic "field defects" in histologically normal tissues proximal to tumours [10, 11, 12]. These mechanisms motivate an exploratory investigation focused not only on overt tumour-associated changes, but also on more subtle alterations in Normal and Adjacent samples, which are central to the biological hypothesis of this thesis.

This section therefore applies a unified exploratory framework to each dataset (GSE69914 [3.3.1], GSE225845 [3.3.2], GSE287331 [3.3.3]), systematically examining data quality, global methylation distributions, sample-level summaries, CpG-wise variability and instability, correlation structure, dimensionality reduction, and genomic context coverage. The use of a consistent analytical protocol enables direct comparison of intra-dataset behaviour while preserving cohort-specific characteristics.

3.3.1 Dataset GSE69914

Dataset overview GSE69914 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium HumanMethylation450 BeadChip, providing genome-wide coverage of approximately 4.8×10^5 CpG loci. The dataset comprises samples spanning the full Normal–Adjacent–Tumor spectrum that motivates the biological framework of this thesis.

An overview of the sample composition and CpG coverage is reported in Table 3.1, including the distribution across tissue classes and hereditary-risk subgroups.

A limited subset of samples is associated with germline *BRCA1* mutations. *BRCA1*, in particular, encodes a key DNA repair protein, and pathogenic variants are known to substantially increase lifetime breast and ovarian cancer risk [13]. Multiple studies have shown that BRCA-associated breast tissues exhibit distinct baseline epigenetic profiles, reflecting inherited genomic instability rather than sporadic tumorigenesis [14, 15].

Since the primary objective of this thesis is to characterise early, field defect DNA methylation alterations along the Normal–Adjacent axis in sporadic breast cancer, samples carrying *BRCA1* mutations were excluded from the core exploratory and

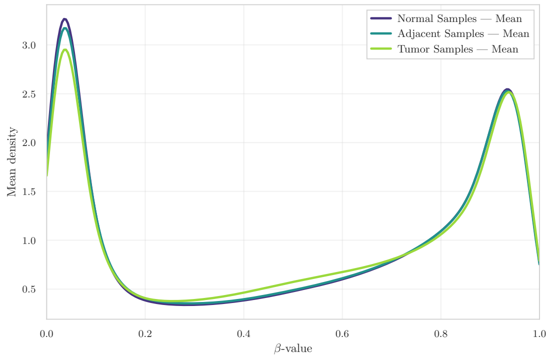


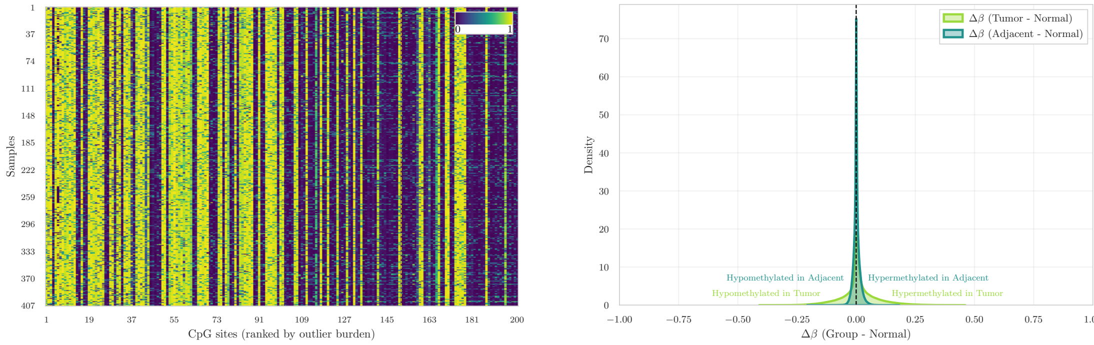
Figure 3.1: Group-wise mean β-value density in GSE69914.

preprocessing pipeline. Their inclusion would introduce a confounding hereditary epigenetic signal, potentially obscuring subtle methylation shifts in non-mutated normal and adjacent tissues.

Global methylation distributions To characterise the overall methylation landscape of the dataset, the distribution of β-values (fractional methylation in [0,1]) was first examined. The group-wise mean density curve (Figure 3.1) displays the characteristic *bimodal* profile of Illumina 450k arrays, with peaks near unmethylated ($\beta \approx 0$) and fully methylated ($\beta \approx 1$) CpG sites. This pattern reflects the underlying biology of CpG regulation, where many loci tend to be either transcriptionally active (hypomethylated) or repressed (hypermethylated) [11, 16].

When comparing tissue groups, a clear gradient emerges. Tumor samples show a slightly flatter high-β peak and a broader low-β tail, consistent with the well-described phenomenon of global hypomethylation and increased heterogeneity in cancer [10]. Adjacent samples lie between Normal and Tumor, suggesting early epigenetic drift and subtle field defects occurring in histologically non-neoplastic tissue [12].

CpG-level instability and recurrent outliers To characterise locus-specific instability, the 200 CpG sites with the highest outlier burden across samples were examined. The heatmap of the top outlier loci (Figure 3.2a) indicates that epigenetic disruption is *not uniform*: specific CpG sites and specific samples display extreme deviations, rather than a diffuse genome-wide shift. Moreover, both directions of alteration are present — focal hypermethylation and focal hypomethylation. In cancer biology, hypermethylation can silence tumor-suppressor regions, whereas hypomethylation can derepress oncogenic pathways and weaken genomic stability, reflecting the classic "too much and too little methylation" behaviour of tumor genomes [10].



(a) Top outlier CpG loci.

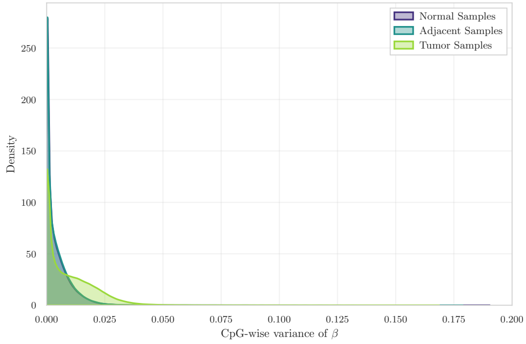
(b) $\Delta\beta$ distributions (Tumor/Adjacent − Normal).

Figure 3.2: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE69914.

Complementary $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Figure 3.2b) show a clear shift toward hypomethylation in Tumor samples and a subtler but detectable drift in Adjacent tissues. This provides evidence that epigenetic alterations emerge early in histologically normal tissue located near the tumor, supporting the field-cancerization model [12].

Variance and correlation structure To quantify within-group epigenetic variability, the distribution of CpG-wise variance across samples was examined. The density curves (Figure 3.3a) show that Tumor samples have markedly higher variance, reflecting increased epigenetic instability. In contrast, Normal and Adjacent samples display almost identical variance distributions that are tightly concentrated near zero. This indicates that, at the level of CpG-wise variability, Adjacent tissues do not yet exhibit detectable divergence from Normal samples.

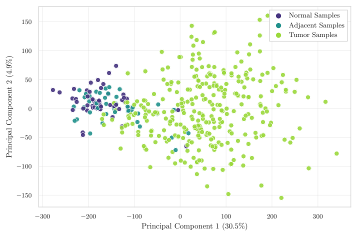
A complementary perspective is provided by the sample correlation heatmap (Figure 3.3b), which shows the expected high level of pairwise similarity across all samples. This behaviour is typical of genome-wide DNA methylation data, where a large fraction of CpG sites is stable across individuals, resulting in consistently strong correlations [7]. Importantly, the heatmap also reveals that all samples are highly mutually correlated, with no sharp blocks or boundaries separating the three tissue labels. This confirms that global methylation structure is largely conserved across Normal, Adjacent and Tumor tissues, and that group differences are too subtle to be detected through sample–sample correlation alone.



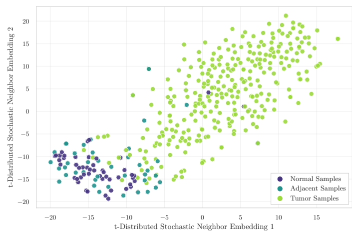
(a) CpG-wise variance distributions.

(b) Sample correlation heatmap.

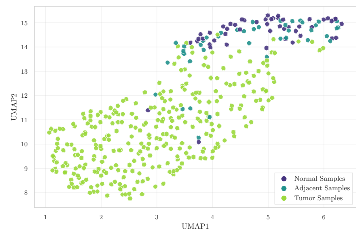
Figure 3.3: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE69914.



(a) PCA trustworthiness = 0.869.



(b) t-SNE trustworthiness = 0.888.

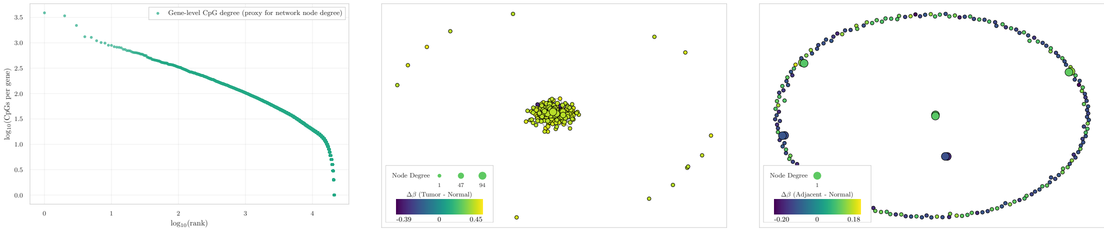


(c) UMAP trustworthiness = 0.874.

Figure 3.4: Low-dimensional embeddings samples in GSE69914.

Low-dimensional embeddings Dimensionality reduction was applied using Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP) to visualise global similarities among samples (Figure 3.4). Across all three methods, Normal and Adjacent samples show substantial overlap, with no clear separation between these two groups. Tumor samples appear more dispersed, but only mildly shifted relative to the Normal/Adjacent cluster, and no sharp cluster boundaries emerge in any embedding. The trustworthiness scores are similar across embeddings (0.869–0.888), indicating that local neighbourhood relationships are largely preserved in the low-dimensional projections.

These projections indicate that global methylation patterns alone do not provide strong discriminative structure among the three tissue states. This suggests that group differences are subtle at the whole-methylome level and are more effectively captured through locus-specific analyses rather than through global unsupervised embeddings.



(a) Gene-level CpG degree distribution.

(b) DNB-like network: Tumor vs Normal.

(c) DNB-like network: Adjacent vs Normal.

Figure 3.5: Dynamic-network proxy visualizations in GSE69914.

Genome annotation coverage CpG probes were annotated using the official *Infinium MethylationEPIC v1.0 B5* [17] manifest file, the standard Illumina resource containing genomic coordinates, probe type, CpG island context (Island, Shore, Shelf, Open Sea), and gene-level annotations. Using this manifest, the dataset exhibits the expected non-uniform genomic distribution of CpG contexts: the largest fraction of probes maps to open-sea or non-island regions, followed by CpG islands, then north and south shores, and finally north and south shelves. This ordering mirrors the characteristic architecture of the HM450 arrays reported in the literature [18, 19]. The coverage observed is therefore consistent with the known genomic composition of the Illumina methylation arrays, supporting correct manifest alignment.

Dynamic-network proxies Although this analysis does not follow the full mathematical formalism of Dynamic Network Biomarkers (DNBs), the degree-based and network-level (Figure 3.5) provide a qualitative view of how methylation perturbations may organize at the network level. DNB theory, originally formulated for gene-expression and regulatory networks, predicts that near a critical transition a subset of components displays increased internal fluctuations and coordinated behaviour [20, 21, 22]. Here, this conceptual framework is adapted to CpG-level methylation patterns by examining the structure of CpG–gene degree distributions and the organization of $\Delta\beta$-driven network layouts. This approach enables the exploration of whether tumour-associated instability manifests as locally coordinated methylation shifts at the network level, even without implementing the complete DNB pipeline.

The degree–rank plot (Figure 3.5a) displays a heavy-tailed distribution of CpG counts per gene, consistent with the heterogeneous hub–periphery structure typically observed in biological systems. In the Tumor vs Normal comparison (Figure 3.5b), a compact nucleus of nodes with moderately high degree is observed, accompanied by larger $\Delta\beta$ deviations. This pattern is consistent with a more perturbed and locally coordinated subset of CpG/gene units in tumor tissue, qualitatively compatible with increased network instability. In contrast, the Adjacent vs Normal network (Figure 3.5c) does not exhibit a comparable hub-like concentration and shows smaller, more diffuse deviations, suggesting that the adjacent tissue reflects only early or weakly organized perturbations.

Although these observations do not satisfy the full mathematical criteria for identifying a DNB module, they are conceptually consistent with the interpretation that tumor tissue occupies a more unstable network regime, whereas adjacent tissue remains closer to a stable (pre-transition) configuration.

Overall interpretation The exploratory analysis of GSE69914 reveals a consistent but overall subtle separation between the three tissue states. Normal and Adjacent samples exhibit highly similar global methylation profiles, variance distributions, correlation structure, and low-dimensional embeddings, indicating that large-scale methylome organisation remains largely conserved in Adjacent tissue. Across these global summaries, the two groups are so intermixed that no clear or marked distinction emerges between Normal and Adjacent samples when using whole-methylome descriptors. Tumor samples display the expected increase in heterogeneity and a shift toward hypomethylation; however, these differences remain modest at the whole-methylome level and do not produce clear separation in unsupervised embeddings.

Locus-specific analyses, however, highlight focused patterns of disruption. Outlier CpGs exhibit both hyper- and hypomethylation events, and $\Delta\beta$ distributions reveal a distinct hypomethylation bias in Tumor samples together with a milder but detectable shift in Adjacent tissue. Dynamic-network proxies further indicate a more compact and perturbed core in Tumor samples, whereas Adjacent tissue appears more diffuse and weakly structured.

Overall, GSE69914 constitutes a technically clean dataset with well-structured methylation patterns and biologically interpretable deviations. The results indicate that tumour-related alterations are primarily localized at specific CpG loci rather than reflected in global methylome reorganization. Within this framework, the Adjacent tissue exhibits subtle yet measurable deviations from the Normal baseline, consistent with previously described field-effect patterns in breast cancer epigenomics.

3.3.2 Dataset GSE225845

Dataset overview GSE225845 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium MethylationEPIC BeadChip (850K), providing genome-wide coverage of more than 8.5×10^5 CpG loci. The dataset is part of the

Table 3.2: Overview of the GSE225845 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor
750,426	477	113	140	224

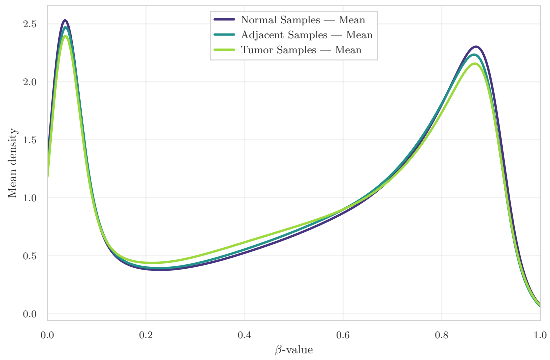


Figure 3.6: Group-wise mean β-value density in GSE225845.

NCI-Maryland Breast Cancer Cohort and includes samples spanning the Normal–Adjacent–Tumor spectrum that motivates the biological framework of this thesis.

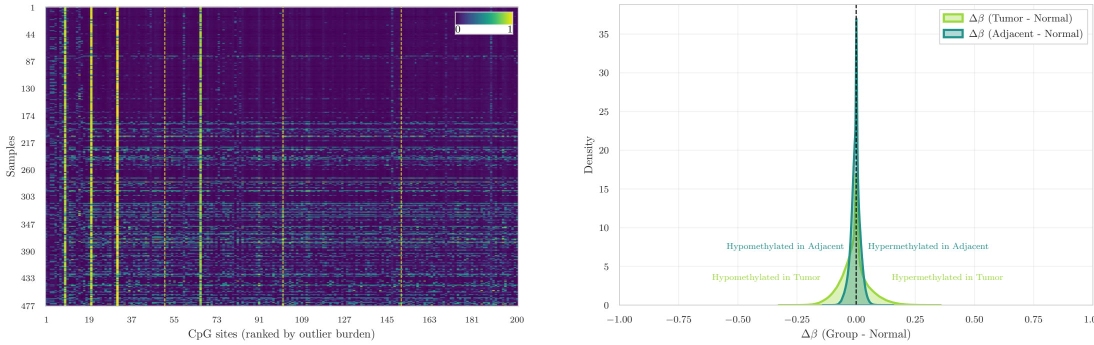
An overview of the sample composition and CpG coverage after cohort harmonization is reported in Table 3.2, including the distribution across tissue classes.

The dataset further provides detailed demographic and clinical metadata (e.g., age at surgery, race, sex, and tissue descriptors), enabling controlled downstream analyses when required.

Global methylation distributions The group-wise mean β-value densities (Figure 3.6) display the expected bimodal configuration of EPIC arrays. As observed in GSE69914, the three tissue groups show a high degree of overlap across the entire β range.

Normal and Adjacent samples exhibit nearly indistinguishable density profiles. Tumor samples present only minimal deviations, characterised by a slight increase in density at low β values and a marginal attenuation of the high-β peak. At the whole-methylome level, no marked global redistribution of methylation levels is evident.

CpG-level instability and recurrent outliers The heatmap of the top outlier CpG loci (Figure 3.7a) indicates that instability within this subset is predominantly directional. Recurrent outlier events are largely associated with very low β values, consistent with focal hypomethylation affecting a restricted set of CpG sites. The perturbation pattern therefore appears concentrated rather than diffuse, suggesting localized instability rather than a genome-wide redistribution.



(a) Top outlier CpG loci.

(b) $\Delta\beta$ distributions (Tumor/Adjacent − Normal).

Figure 3.7: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE225845.

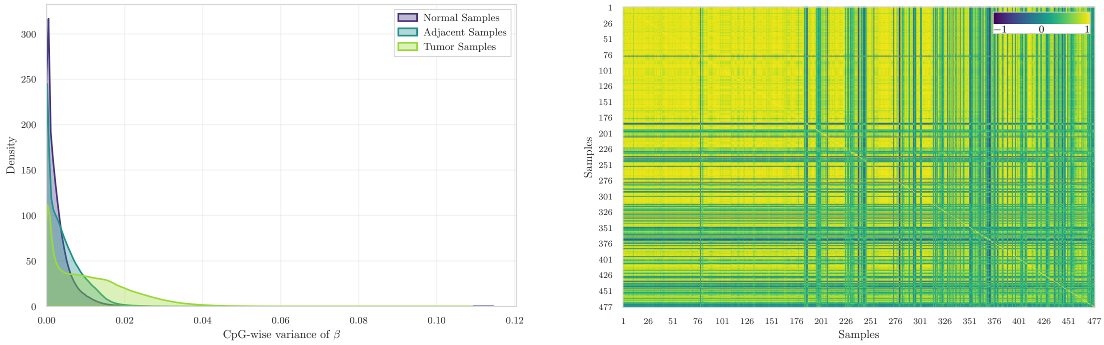
The corresponding $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Figure 3.7b) are sharply centered around zero, indicating that most loci remain stable. However, both comparisons exhibit a heavier negative tail relative to the positive side. Tumor samples display the most extended negative tail, consistent with stronger hypomethylation shifts, whereas Adjacent tissue shows a milder but still detectable asymmetry toward $\Delta\beta < 0$.

Variance and correlation structure The CpG-wise variance distributions (Figure 3.8a) indicate clear differences in variability across tissue groups. Normal samples exhibit highly concentrated variance values near zero, whereas Tumor samples display a substantially longer right tail, reflecting an increased proportion of CpG loci with elevated variability. Adjacent samples show a broader distribution than Normal tissues, with partial overlap with both Normal and Tumor profiles, but without forming a strictly intermediate pattern.

The sample correlation heatmap (Figure 3.8b) shows uniformly positive correlations across the dataset. Most pairwise values lie within a narrow high-correlation range, and no distinct block structures separating tissue classes are observed. This pattern indicates preservation of global methylation structure, with group differences emerging primarily from locus-specific variability rather than from large-scale shifts in overall methylome organization.

Low-dimensional embeddings Dimensionality reduction was applied using PCA, t-SNE, and UMAP to visualise global similarities among samples (Figure 3.9).

In PCA space, the three tissue groups largely overlap. Tumor samples show a slightly broader dispersion, particularly along the second principal component, but



(a) CpG-wise variance distributions.

(b) Sample correlation heatmap.

Figure 3.8: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE225845.

(a) PCA trustworthiness = 0.881.

(b) t-SNE trustworthiness = 0.939.

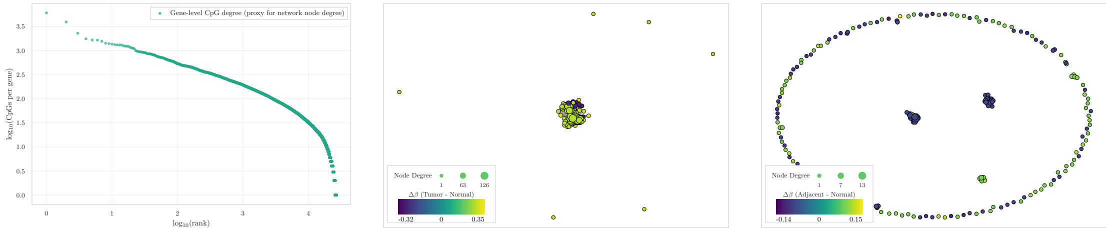
(c) UMAP trustworthiness = 0.894.

Figure 3.9: Low-dimensional embeddings samples in GSE225845.

no clear linear separation is observed.

Nonlinear embeddings reveal additional structure. In t-SNE space, Tumor samples occupy a more extended region and appear more dispersed, whereas Normal and Adjacent samples remain partially intermixed. Several Tumor samples are positioned at the periphery of the embedding, consistent with increased heterogeneity.

The UMAP projection exhibits a similar configuration. Tumor samples span a wider area and extend toward higher values of the second UMAP axis, while Normal and Adjacent samples form relatively compact yet overlapping regions. Trustworthiness scores range from 0.881 to 0.939 across embeddings, indicating good preservation of local neighbourhood structure, particularly in the nonlinear projections. Despite differences in spread, none of the methods yields sharply separated or isolated clusters.



(a) Gene-level CpG degree distribution.

(b) DNB-like network: Tumor vs Normal.

(c) DNB-like network: Adjacent vs Normal.

Figure 3.10: Dynamic-network proxy visualizations in GSE225845.

Genome annotation coverage Consistent with the architecture of the EPIC platform, the genomic distribution of annotated CpG contexts is markedly unbalanced. The largest fraction of probes maps to open-sea or non-island regions, which constitute the dominant category. CpG islands represent the second most abundant class, followed by north and south shores in comparable proportions, whereas north and south shelves account for the smallest fractions.

This annotation profile aligns with the expected genomic composition of the EPIC array and supports correct manifest integration.

Dynamic-network proxies The degree–rank distribution (Figure 3.10a) exhibits a heavy-tailed profile, with a limited number of genes associated with many CpGs and a long tail of low-degree nodes. Such right-skewed degree structures are characteristic of heterogeneous biological networks [23].

In the Tumor vs Normal layout (Figure 3.10b), nodes aggregate into a dense and highly compact central cluster, with only a small number of isolated peripheral points. This geometry indicates that the largest Tumor–Normal $\Delta\beta$ deviations are concentrated within a cohesive subset of CpG–gene units rather than broadly distributed across the network.

In contrast, the Adjacent vs Normal layout (Figure 3.10c) displays a markedly different configuration. While a few small central aggregates are visible, most nodes are arranged along a wide annular structure, forming a pronounced ring-like embedding. This organisation reflects weaker and more spatially dispersed deviations, without the central concentration observed in the Tumor comparison.

Altogether, the proxy visualisations indicate that Tumor–Normal contrasts generate a compact and locally coherent network signal, whereas Adjacent–Normal differences remain diffuse and geometrically fragmented.

Overall interpretation The exploratory analysis of GSE225845 indicates that, at the whole-methylome level, the three tissue states share highly similar global

Table 3.3: Overview of the GSE287331 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor	OQ	CUB
866,552	446	182	60	69	67	68

methylation profiles. Group-wise mean β-value densities are nearly superimposed, and both correlation structure and linear embeddings show substantial overlap. These observations indicate preservation of large-scale methylome organisation across Normal, Adjacent, and Tumor tissues, without evidence of pronounced genome-wide redistribution of methylation levels.

Locus-focused analyses reveal more structured differences. Among the 200 CpGs with the highest outlier burden, instability is predominantly directional and characterised by focal hypomethylation. Variance distributions further show increased dispersion in Tumor samples, while Adjacent tissues exhibit broader variability than Normal without forming a strictly intermediate pattern.

Nonlinear embeddings (t-SNE and UMAP) emphasise differences in spatial dispersion rather than discrete cluster separation. Normal and Adjacent samples occupy relatively compact regions, whereas Tumor samples extend over a wider area, reflecting increased heterogeneity. Dynamic-network proxies show that Tumor–Normal contrasts generate a dense central cluster of perturbed CpG–gene units, while Adjacent–Normal differences are distributed along a broader, ring-like structure without strong central concentration.

Despite the larger sample size of this cohort, which provides stable estimates of global methylation structure, the separation between Normal and Adjacent tissues remains modest. Differences between these two groups are therefore primarily focal and do not manifest as large-scale structural shifts.

Overall, GSE225845 represents a globally homogeneous dataset in which tissue-state differences become apparent only when examined through locus-specific perturbations, variability patterns, and small-scale network organisation.

3.3.3 Dataset GSE287331

Dataset overview GSE287331 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium MethylationEPIC v1.0 BeadChip. The dataset is organised along a tumor proximity axis (TPxA) spanning multiple anatomical and pathological contexts.

Five tissue categories are represented: healthy donated breast (HDB), contralateral unaffected breast (CUB), ipsilateral opposite quadrant (OQ), adjacent normal tissue (AN), and tumor (TU). An overview of the full cohort composition is reported in Table 3.3.

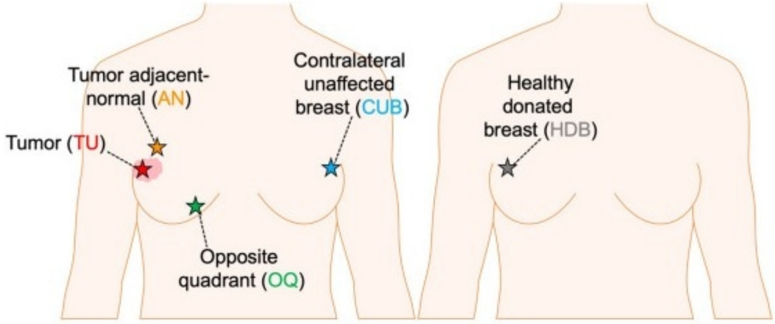


Figure 3.11: Schematic representation of the five tissue categories collected along the tumor proximity axis (TPxA), reproduced from [24].

For cross-dataset consistency, only three biologically aligned classes are retained for downstream analyses: HDB (treated as Normal baseline), AN (Adjacent), and TU (Tumor). OQ and CUB samples are excluded, as they represent intermediate benign tissues along the proximity axis and are not directly comparable with the three-class framework adopted in the other cohorts.

Global methylation distributions The group-wise mean β-value density curves (Figure 3.12) display the expected bimodal configuration of Illumina methylation arrays. A dominant peak is observed at high methylation levels, accompanied by a smaller peak consistent with standard EPIC methylation profiles [25].

Although the overall shapes remain comparable across tissue groups, a clear ordering is visible at the high-β peak. Normal samples exhibit the highest density, Adjacent samples show a slightly attenuated peak, and Tumor samples present the lowest amplitude together with a broader distribution extending toward intermediate β values. This progressive reduction in the fully methylated peak is consistent with increasing methylation variability along the Normal–Adjacent–Tumor axis.

CpG-level instability and recurrent outliers The heatmap of the top outlier CpG loci (Figure 3.13a) shows that epigenetic instability is not uniformly distributed. Alterations concentrate at specific CpG sites and within specific samples, forming vertical and horizontal bands rather than a diffuse genome-wide pattern. Both directions of deviation are observed, with focal hypermethylation and focal hypomethylation present within the selected loci [10].

The corresponding $\Delta\beta$ distributions (Figure 3.13b) highlight distinct profiles for the two contrasts. The Tumor–Normal curve is broader and displays a higher density of small negative shifts, indicating widespread but moderate hypomethylation in Tumor samples. In contrast, the Adjacent–Normal distribution is more sharply centered around zero but exhibits a comparatively more pronounced negative tail,

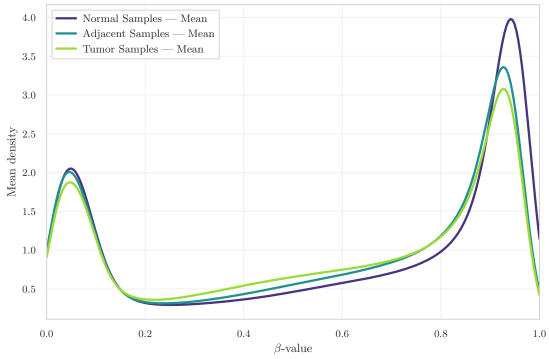
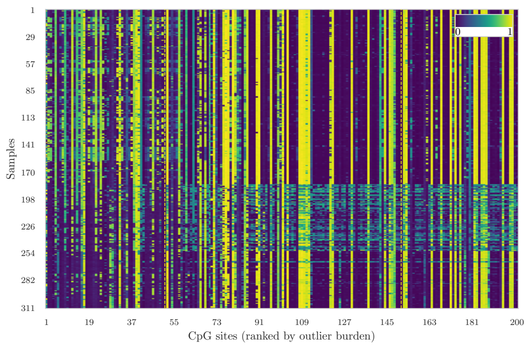
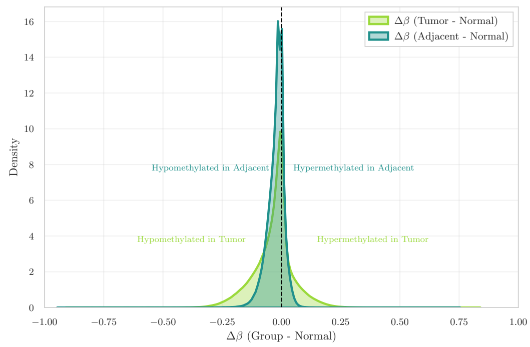


Figure 3.12: Group-wise mean β-value density in GSE287331.



(a) Top outlier CpG loci.



(b) $\Delta\beta$ distributions (Tumor/Adjacent − Normal).

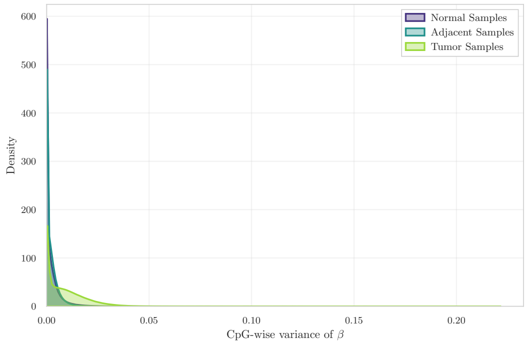
Figure 3.13: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE287331.

reflecting fewer yet larger hypomethylated deviations. Overall, hypomethylation represents the dominant direction of change in both comparisons.

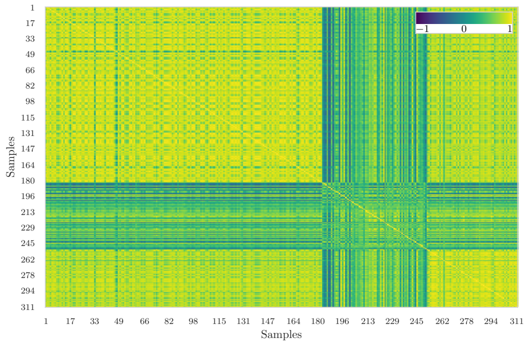
Variance and correlation structure The CpG-wise variance density curves (Figure 3.14a) show that Normal and Adjacent samples are nearly superimposed, with both groups exhibiting extremely low and tightly concentrated variance values. Tumor samples display a slightly broader right tail, indicating modestly increased genome-wide variability; however, the overall distribution remains sharply peaked near zero.

At the level of CpG-wise variance, Adjacent tissue remains closely aligned with the Normal baseline, whereas Tumor samples show only a mild elevation in variability.

A complementary perspective is provided by the sample correlation heatmap



(a) CpG-wise variance distributions.



(b) Sample correlation heatmap.

Figure 3.14: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE287331.

(Figure 3.14b). Correlations are uniformly high across the cohort, consistent with the predominance of stable CpG loci in genome-wide methylation data [7]. Although tissue labels are not overlaid on the matrix, localized blocks of higher similarity are visible, indicating structured relationships within subsets of samples rather than abrupt global separation.

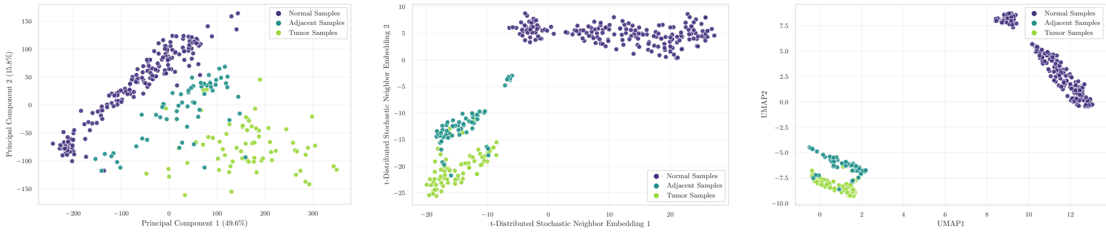
Low-dimensional embeddings Dimensionality reduction using PCA, t-SNE, and UMAP was applied to visualise the global organisation of methylation profiles (Figure 3.15).

Across all three embeddings, a consistent and well-defined structure is observed. Normal and Adjacent samples form two clearly separated clusters. The separation is visible in PCA space, becomes more pronounced in t-SNE, and appears as fully detached manifolds in the UMAP projection.

Tumor samples occupy a distinct and more dispersed region of the embedding space, reflecting increased epigenetic heterogeneity. Trustworthiness scores are uniformly high (0.961–0.980), indicating excellent preservation of local neighbourhood structure in all low-dimensional projections. No substantial overlap between Tumor and the other two groups is observed in any of the projections, indicating strong intrinsic group structure in this cohort.

Genome annotation coverage The genomic distribution of annotated CpG contexts is dominated by Open Sea regions, followed by CpG Islands, then Shores, with Shelves representing the smallest fractions.

This ordering is consistent with the established architecture of EPIC arrays [18, 19].

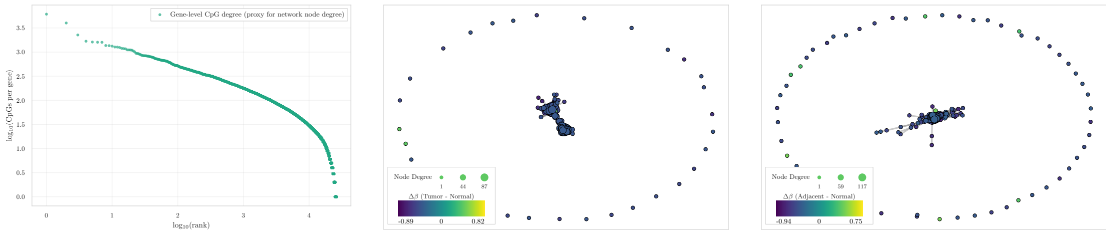


(a) PCA trustworthiness = 0.961.

(b) t-SNE trustworthiness = 0.980.

(c) UMAP trustworthiness = 0.973.

Figure 3.15: Low-dimensional embeddings of GSE287331 samples.



(a) Gene-level CpG degree distribution.

(b) DNB-like network: Tumor vs Normal.

(c) DNB-like network: Adjacent vs Normal.

Figure 3.16: Dynamic-network proxy visualizations for GSE287331.

Dynamic-network proxies The degree–rank curve (Figure 3.16a) shows a heavy-tailed distribution of CpG counts per gene, consistent with a hub–periphery architecture typical of biological networks.

Both $\Delta\beta$-based network layouts — Tumor vs Normal and Adjacent vs Normal (Figure 3.16b, Figure 3.16c) — share a similar overall structure: a dense and compact central block of nodes connected by short-range edges, surrounded by a ring of peripheral, weakly connected nodes. This geometry reflects the underlying degree distribution and indicates that most CpG–gene units contribute only minimally to group-level differences.

Despite this shared organisation, the Tumor vs Normal comparison exhibits a more pronounced central cluster with larger $\Delta\beta$ magnitudes, suggesting that a restricted subset of CpG–gene units undergoes locally coordinated perturbations. In contrast, the Adjacent vs Normal layout preserves the same global shape but shows weaker deviations, indicating only mild coordination of methylation changes.

Overall interpretation The exploratory analysis of GSE287331 indicates that, at the whole-methylome level, the three tissue states retain a broadly preserved global methylation architecture. Group-wise mean β-value densities follow the typical bimodal EPIC profile with only a subtle gradient (Normal > Adjacent > Tumor), and both CpG-wise variance distributions and sample correlation structure show high overall similarity across tissues, with only mild variability increases in Tumor samples.

In contrast to the previous cohorts, low-dimensional embeddings reveal a clear and consistent separation of all three groups. Normal and Adjacent samples form distinct clusters in PCA, with the separation further strengthened in t-SNE and UMAP, where the groups appear as detached manifolds. Tumor samples occupy a separate and more diffuse region, reflecting increased epigenetic heterogeneity. These projections indicate that, despite similar first- and second-order global summaries, multivariate structure in this dataset captures strong group-level organisation.

Locus-specific analyses reveal focused disruptions. Tumor samples exhibit widespread mild hypomethylation, whereas Adjacent tissue shows fewer but larger-magnitude deviations. Dynamic-network proxies further indicate a compact and perturbed central core in the Tumor comparison, while Adjacent–Normal differences display weaker coordination, consistent with early-stage methylation alterations.

Overall, GSE287331 emerges as a structurally coherent and biologically informative dataset in which Normal–Adjacent–Tumor separation is markedly more pronounced than in the other cohorts, while global methylation architecture remains broadly conserved.

3.4 Inter Dataset Exploration and Comparison

The objective of this section is not to merge the datasets—an approach that would be biologically and technically inappropriate due to differences in array platforms (HM450 vs. EPIC), preprocessing pipelines—but rather to compare their internal behaviours. Such a cross-dataset perspective enables the assessment of whether the early epigenetic alterations identified in the intra-dataset analyses are *reproducible across independent cohorts*, thereby providing support for the central biological premise of this thesis: the detection of early methylation drift in histologically normal tissues and the evaluation of its potential contribution to tumour initiation.

3.4.1 Analytical Framework

Across all three datasets, the intra-dataset analyses revealed three highly consistent phenomena:

- Normal and Adjacent tissues are globally similar, with nearly superimposed β-value distributions and comparable CpG-wise variance profiles.

Table 3.4: Cross-dataset comparison of Normal (N) vs Adjacent (A) contrasts using global and instability metrics.

Dataset	**Mean $\lvert\Delta\beta\rvert$ (A–N)**	**Var Ratio (A/N)**	**Outlier Ratio (A/N)**	**Silhouette (A/N)**
GSE69914	0.009	0.988	1.353	0.003
GSE225845	0.015	0.972	2.790	0.042
GSE287331	0.031	0.930	13.329	0.344

- Tumour samples show increased heterogeneity and a clear bias toward hypomethylation, although the magnitude of this shift varies across datasets.
- Adjacent tissues exhibit early methylation drift, detectable through $\Delta\beta$ distributions and outlier profiles even when global metrics remain highly similar to Normal.

These patterns are consistent with the expected biological progression from Normal to Tumour, in which epigenetic deregulation emerges gradually and affects normal-appearing tissue surrounding the tumour. The inter-dataset analysis therefore evaluates the *directional consistency* of these effects rather than aiming at strict quantitative comparability.

3.4.2 Cross-Dataset Comparative Metric

To quantify the epigenetic deviation of Adjacent tissue from the Normal methylome across the three cohorts, a set of global and instability-oriented metrics was computed independently within each dataset for the Adjacent versus Normal contrast.

All metrics reported in Table 3.4 were computed using the complete CpG set available in each dataset, rather than the cross-platform intersection. The resulting values therefore reflect cohort-specific dimensionality and internal variance structure.

Collectively, these metrics indicate the presence of early epigenetic drift in histologically non-tumour tissue. Across cohorts, increasing mean $|\Delta\beta|$, progressive variance imbalance, elevated outlier ratios, and improved silhouette separation consistently support a directional shift from Normal to Adjacent states, in agreement with the field-defect framework described by Teschendorff et [12] and reinforced in other tissue contexts.

Mean absolute methylation difference For each CpG, the absolute mean difference between Adjacent and Normal was defined as

$$|\Delta\beta_i| = \left|\beta_i^{(A)} - \beta_i^{(N)}\right|. \tag{3.2}$$

The cross-cohort pattern reveals progressively larger locus-specific shifts from GSE69914 to GSE225845 and GSE287331. Although the absolute magnitude of the effect remains modest, it is consistent in direction across cohorts. This monotonic increase reflects a strengthening early deviation in Adjacent tissues. Comparable locus-wise methylation drifts in histologically normal tissue have been reported in independent systems, such as colonic mucosa adjacent to adenomas [26], supporting the interpretation of these alterations as early manifestations of field cancerization.

Global variance ratio For each sample s, the global methylation variance was computed across all CpG sites: $v_s = \mathrm{Var}(\beta_{s,1}, \beta_{s,2}, \ldots, \beta_{s,m})$, where m denotes the number of CpGs. Group-wise mean variances were then defined as

$$\overline{v}^{(N)} = \frac{1}{|N|} \sum_{s \in N} v_s, \qquad \overline{v}^{(A)} = \frac{1}{|A|} \sum_{s \in A} v_s, \tag{3.3}$$

and the reported metric corresponds to their ratio:

$$\frac{\mathrm{Var(A)}}{\mathrm{Var(N)}} = \frac{\overline{v}^{(A)}}{\overline{v}^{(N)}}. \tag{3.4}$$

Across cohorts, this ratio remained consistently close to unity, indicating that Adjacent tissue does not exhibit diffuse genome-wide instability. Such behaviour suggests that early pre-neoplastic alterations are predominantly *focal* rather than global, with only specific CpG sites displaying abnormal dispersion while overall variance remains largely preserved [26].

This interpretation is consistent with contemporary models of early tumourigenesis, in which subtle and locally restricted perturbations accumulate prior to large-scale chromatin deregulation and overt neoplastic transformation [27].

Outlier burden ratio For each sample, the outlier burden is defined as the number of CpG sites whose methylation deviates from the Normal reference distribution by more than three times the Normal interquartile range:

$$\mathrm{outlier}_s = \# \left\{ i \; : \; \left| \beta_{i,s} - \tilde{\beta}_i^{(N)} \right| > 3 \cdot IQR_i^{(N)} \right\}, \tag{3.5}$$

where $\tilde{\beta}_i^{(N)}$ and $IQR_i^{(N)}$ denote the median and interquartile range of CpG i across Normal samples.

The reported A/N ratio corresponds to the mean outlier burden in Adjacent tissue divided by the mean burden in Normal tissue. Across cohorts, this metric exhibits the most pronounced cross-dataset gradient, with progressively stronger enrichment of outlier events in Adjacent tissue. Such behaviour mirrors evidence that increased local methylation variability in histologically normal tissue represents a sensitive marker of early instability [26], and provides quantitative support for the presence of marked field effects in GSE287331.

Silhouette score For each sample s, let $a(s)$ denote the mean distance from s to all other samples within the same group (cohesion), and let $b(s)$ denote the smallest mean distance from s to samples in the alternative group (separation). The silhouette coefficient for sample s is defined as

$$\mathrm{sil}(s) = \frac{b(s) - a(s)}{\max\{a(s),\, b(s)\}}, \tag{3.6}$$

and the reported score corresponds to the mean silhouette computed across all Normal and Adjacent samples.

Higher values indicate clearer multivariate separation between Normal and Adjacent states, whereas values near zero reflect substantial overlap. Across cohorts, the silhouette analysis reveals limited separation in GSE69914 and GSE225845, consistent with minimal global displacement in multivariate space. In contrast, GSE287331 exhibits partial separation, indicating that the epigenetic landscape of Adjacent tissue is measurably shifted relative to Normal. This behaviour aligns with mechanistic models in which epigenetic dysregulation constitutes one of the earliest events in tumour initiation [27], preceding morphological transformation.

Collectively, the set of metrics delineates a consistent Normal–Adjacent gradient across datasets. Early epigenetic drift is detectable in all cohorts, while its magnitude, focality, and multivariate visibility progressively intensify from GSE69914 to GSE287331. Importantly, the combination of stable global variance with pronounced outlier enrichment supports a model in which early pre-neoplastic changes arise through *localised* perturbations, a hallmark of field cancerization [26, 12].

3.4.3 Visual Comparative Analysis

To complement the quantitative summary reported in Table 3.4, this section provides a visual characterisation of the Normal–Adjacent contrast across the three cohorts. All analyses are performed on the three-way intersection of 326,330 CpG sites, ensuring maximal comparability across platforms (HM450 and EPIC) and preprocessing pipelines [18]. The objective is not to achieve numerical concordance at the single-CpG level—a notoriously difficult task due to platform differences, probe-design effects, and cohort-specific processing [19, 28]—but rather to evaluate whether the *shape*, *direction*, and *global behaviour* of early methylation drift remain coherent across studies.

Global $\Delta\beta$ distributions The overlaid $\Delta\beta$ (Adjacent–Normal) density curves across the three cohorts are shown in Figure 3.17. All distributions are sharply centred around zero, indicating that the N–A methylation drift remains globally subtle in each dataset, consistent with the focal and low-amplitude nature of early

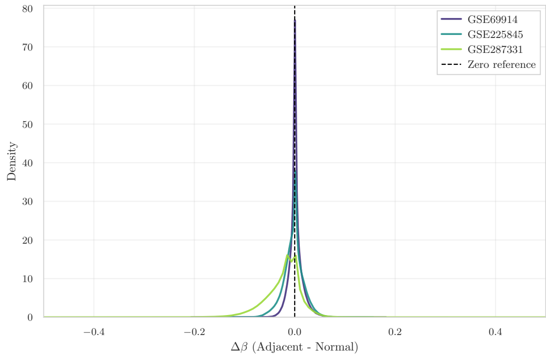


Figure 3.17: Overlay of $\Delta\beta$ (Adjacent–Normal) density curves across the three datasets, computed on the 326,330 CpG sites shared by all cohorts.

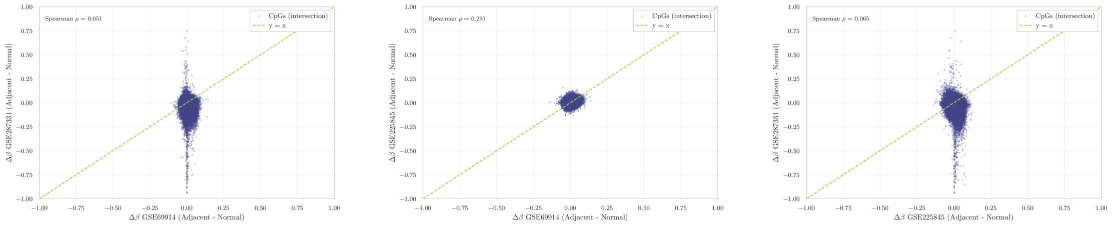
epigenetic alterations described in pre-neoplastic tissues [12, 29]. Despite differences in dispersion—reflecting platform-dependent noise, probe-type composition, and study-specific variability [16, 19]—the three curves exhibit a highly similar shape. This pattern indicates that early epigenetic deviations in Adjacent tissue, although small in magnitude, exhibit reproducible global distributional features across independent cohorts, rather than exact concordance at the single-CpG level.

Pairwise $\Delta\beta$ concordance across datasets To assess whether locus-specific A–N shifts replicate across cohorts, pairwise scatterplots are shown for all dataset pairs (Figure 3.18a, Figure 3.18b, Figure 3.18c). Each point corresponds to a CpG in the three-way intersection; axes represent the mean $\Delta\beta$ (A–N) computed independently within each dataset. Across all pairs, Spearman correlations remain weak ($\rho = 0.05$–0.29), consistent with the well-established difficulty of achieving single-CpG reproducibility across independent methylation studies [18, 28, 30].

The comparison between GSE69914 and GSE225845 exhibits mild monotonic agreement, whereas pairs involving GSE287331 show near-zero correlation. These results indicate that while the *global* N–A drift remains consistent across cohorts (as demonstrated by the density curves), *fine-scale, CpG-specific* effects are strongly influenced by platform architecture, batch structure, and study-specific variability [19, 28].

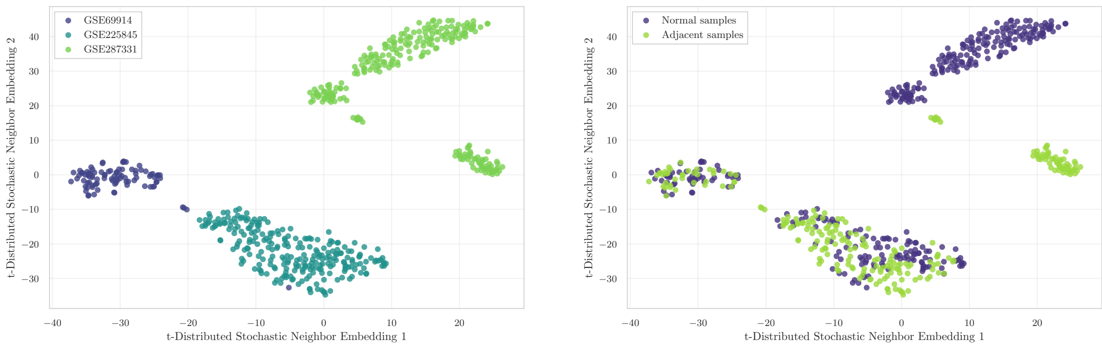
t-SNE embeddings on the shared CpG intersection To investigate the multivariate structure of the N–A contrast across cohorts, t-SNE embeddings are computed on M-values restricted to the shared CpGs. Two complementary visualisations of the embedding are provided in Figure 3.19a (coloured by dataset) and Figure 3.19b (coloured by tissue state).

When coloured by dataset, the embedding exhibits clear clustering by study,



(a) GSE69914-GSE287331, Spearman $\rho \approx 0.051$ **(b)** GSE69914-GSE225845, Spearman $\rho \approx 0.291$ **(c)** GSE225845-GSE287331, Spearman $\rho \approx 0.007$

Figure 3.18: Pairwise $\Delta\beta$ (Adjacent–Normal) scatterplots across the three methylation datasets, computed on the 326,330 shared CpG sites. Spearman ρ values are reported in each panel.



(a) t-SNE coloured by dataset trustworthiness = 0.976 **(b)** t-SNE coloured by tissue state trustworthiness = 0.976

Figure 3.19: t-SNE embeddings computed on the 326,330 CpGs shared across GSE69914, GSE225845, and GSE287331.

reflecting the dominant influence of batch and platform effects in high-dimensional methylation data [28, 30]. This pattern indicates that the global methylation structure is largely driven by inter-cohort variability, supporting the decision not to merge datasets at the raw feature level.

When coloured by tissue state (Normal vs. Adjacent), no global separation across cohorts emerges, as the embedding remains primarily structured by dataset identity. However, within individual study-specific clusters, a state-dependent displacement becomes visible. This separation is particularly evident in GSE287331, where Normal and Adjacent samples form more clearly distinct substructures. This behaviour is consistent with early, localised methylation drift, mirroring the quantitative metrics reported in Table 3.4.

Overall, the visual analyses corroborate the quantitative metrics: early epigenetic drift is detectable in all cohorts, but its magnitude and visibility vary substantially, with GSE287331 exhibiting the clearest field-defect signature. The combination of globally subtle yet directionally consistent $\Delta\beta$ patterns, weak CpG-level concordance, and dataset-specific multivariate structure reinforces the view that the Normal→Adjacent transition reflects genuine early epigenetic deregulation rather than dataset-specific artefacts.

3.5 Synthesis of Exploratory Findings and Pre-Processing Rationale

The exploratory analyses presented in this chapter reveal a consistent picture across all three cohorts: the Normal–Adjacent transition does not manifest as a global reorganisation of the methylome, but as a subtle, focal drift that becomes detectable only through locus-specific perturbations and multivariate displacements. This characteristic makes it particularly sensitive to technical sources of variability — probe-design bias, batch effects, and platform differences — which operate at a comparable or larger scale than the biological signal of interest.

This sensitivity has direct methodological consequences. Raw methylation data, as distributed via GEO, cannot be used directly for downstream modelling without first addressing these technical confounders: doing so would risk attributing platform-specific artefacts to genuine epigenetic alterations, or conversely, suppressing the subtle Normal–Adjacent differences under overly aggressive normalisation. The challenge is therefore not simply to clean the data, but to do so in a way that preserves the biological structure identified here.

Chapter 4 addresses this challenge by developing a structured, platform-aware preprocessing pipeline. Each step is motivated by the specific limitations identified in the present chapter.

Chapter 4

Data preprocessing

4.1 Preprocessing Rationale and Technical Constraints

DNA methylation arrays provide high-dimensional, genome-wide measurements in which biological variability and platform-specific technical artefacts are inevitably superimposed. The exploratory analyses of Chapter 3 revealed a consistent picture across all three cohorts: Normal and Adjacent tissues exhibit nearly identical global methylation distributions, overlapping variance profiles, and indistinguishable sample-level correlation structure. Meaningful differences between tissue states emerge exclusively at focal CpG loci—as quantified by locus-specific $\Delta\beta$ distributions and outlier burden ratios—and remain subtle even at that scale (mean $|\Delta\beta|$ ranging from 0.009 in GSE69914 to 0.031 in GSE287331). This characteristic makes the Normal–Adjacent signal particularly vulnerable to technical confounders, which operate at a comparable or larger scale than the biological effect of interest. Preprocessing must therefore be precise: aggressive enough to remove technical noise, but conservative enough to preserve the subtle epigenetic drift that motivates this thesis.

Three main classes of technical issues must be systematically addressed before any statistical modelling can proceed. First, **probe-design bias** intrinsic to the Infinium platform arises from the coexistence of two chemically distinct probe types (Type I and Type II), which produce inherently different β-value distributions and can distort cross-sample comparability if not corrected [31, 32]. This issue is not uniform across datasets: as will be shown in Section 4.3, GSE69914 and GSE225845 were released with normalization already applied by the original authors, whereas GSE287331 shows clear diagnostic evidence of missing probe-type correction, requiring an additional preprocessing decision with non-trivial consequences for downstream group separation. Second, a subset of **unreliable or biologically**

confounded probes—including cross-reactive loci, SNP-affected sites, non-CpG targets, and sex-chromosome probes—introduces systematic measurement error that can generate spurious associations or mask genuine methylation differences [33, 34]. Third, the **statistical scale of β-values** poses inferential challenges: the bounded [0,1] support and strong mean-dependent heteroscedasticity of β-values violate the assumptions of standard parametric models, motivating transformation to a more statistically well-behaved representation prior to modelling [31].

This chapter addresses these three issues through a structured, platform-aware preprocessing pipeline applied consistently across all cohorts. Section 4.2 describes the general framework and the rationale for each step. Section 4.3 documents the dataset-specific application, focusing on the decisions that deviate from the standard workflow—most notably the handling of probe-type bias in GSE287331, where the tension between technical correction and preservation of biological structure requires explicit justification. Section 4.4 summarises the effect of preprocessing on CpG dimensionality and cross-dataset overlap, providing the basis for the feature selection strategy developed in Chapter 5.

4.2 General Preprocessing Framework

This section describes the general preprocessing framework applied to all three methylation datasets. The pipeline comprises four sequential steps, each addressing a distinct class of technical issue identified in the overview above. Two of these steps—data integrity verification and Infinium probe-design bias correction—require dataset-specific decisions whose rationale and consequences are discussed in Section 4.3. The remaining two steps—technical probe filtering and β-to-M transformation—are applied uniformly across all cohorts following the same protocol.

4.2.1 Structural Integrity and Cohort Harmonization

Prior to any preprocessing operation, each dataset undergoes a structural verification stage to ensure the consistency and completeness of the input data. This step checks for correct alignment between the methylation matrix and its associated phenotype table, verifies sample identifier uniqueness, and confirms the absence of missing metadata fields. Only samples for which both a complete methylation profile and a valid tissue label are available are retained for downstream analysis.

Probe-level and sample-level missingness in the β-value matrix is also assessed at this stage. For two of the three datasets (GSE69914 and GSE225845), the released matrices are fully observed and require no imputation. For GSE287331, a non-negligible fraction of CpG sites exhibits missing values across samples, necessitating a dedicated filtering and imputation strategy consistent with current recommendations for EPIC-array quality control [**Islam2019**, **Mansell2019**]. The dataset-specific treatment of missingness is detailed in Section 3.3.3.

Finally, cohort harmonization is applied where necessary to align tissue labels to the three-class framework (Normal, Adjacent, Tumor) adopted uniformly across datasets. Samples belonging to tissue categories outside this framework are excluded at this stage to ensure cross-dataset comparability.

4.2.2 Probe-Type Bias Diagnostics and Correction

A well-documented source of systematic bias in Illumina Infinium methylation arrays is the coexistence of two chemically distinct probe designs. Type I probes use two bead types per CpG (one for the methylated channel, one for the unmethylated channel), whereas Type II probes use a single bead type and rely on single-base extension with two-color detection. This design difference produces inherently distinct β-value distributions for the two probe types: Type I probes tend to yield more extreme values near 0 and 1, while Type II probes exhibit a broader, intermediate distribution. If uncorrected, this imbalance propagates into downstream statistical analyses and can generate probe-type-specific artefacts in differential methylation testing [**Teschendorff2013**, 35].

The canonical approach to mitigating this bias is Beta-Mixture Quantile normalisation (BMIQ), introduced by Teschendorff et al. [**Teschendorff2013**]. BMIQ models the β-value distribution of each probe type as a three-component Beta mixture—representing unmethylated, hemimethylated, and fully methylated states—and applies a quantile transformation to rescale Type II probe values to match the distributional shape of Type I probes within each sample. The correction is applied independently per sample, so no cross-sample information is introduced.

For each dataset, a systematic diagnostic is performed prior to any correction step, following the procedure described in Teschendorff et al. [**Teschendorff2013**] and Wang et al. [**Wang2015**]: probes are stratified by design type using the official Illumina manifest, and two complementary assessments are carried out—(i) a density comparison of β-value distributions for Type I and Type II probes, and (ii) a quantile–quantile plot of Type II versus Type I quantiles, with the median absolute deviation (MAD) of the Q–Q curve from the identity line as a scalar diagnostic metric. Datasets for which the original authors applied BMIQ or an equivalent normalisation are expected to exhibit near-overlapping distributions and a MAD close to zero; datasets lacking probe-type correction will display a pronounced distributional mismatch and elevated MAD.

The outcome of this diagnostic, and the corrective action taken, differs across cohorts and is reported in full in Section 4.3. Critically, the decision of whether to apply BMIQ is not made mechanically: as discussed in Section 3.3.3, BMIQ correction can attenuate genuine biological structure when group-level differences are expressed as broad distributional shifts, a phenomenon documented in systematic evaluations of EPIC-array normalisation methods [**Welsh2023**, **Liu2016**, 1]. The choice between correcting and preserving the raw data is therefore made on the basis of empirical evidence from the diagnostic plots and quantitative assessment of group separability before and after correction.

4.2.3 Technical Probe Filtering

After addressing probe-design bias, unreliable CpG loci are removed through a systematic filtering procedure based on curated external annotation resources. The objective is to exclude probes whose measured β-values may reflect technical measurement error rather than genuine DNA methylation, thereby reducing noise and improving the reliability of downstream differential analyses.

Four categories of technically problematic probes are targeted:

Cross-reactive and multi-mapping probes. Probes that hybridise to multiple genomic loci produce β-values that cannot be unambiguously attributed to a single CpG site. These are identified using the catalogues of Naeem et al. [**Naeem2014**], Chen et al. [**Chen2013**], Pidsley et al. [**Pidsley2016**], Zhou et al. [**Zhou2017**], and McCartney et al. [**McCartney2016**]. Full descriptions of each resource are provided in Appendix **??**.

SNP-affected probes. Probes whose interrogated CpG dinucleotide, single-base extension site, or probe body overlaps a common genetic variant may measure genotype rather than methylation status, producing population-stratified artefacts in group comparisons [**Zhou2017**, **Pidsley2016**]. These are excluded using the `MASK_snp5` and `MASK_extBase` fields from the Zhou annotation [**Zhou2017**], supplemented by variant-aware lists from Naeem et al. [**Naeem2014**].

Non-CpG probes. Probes targeting non-CpG cytosine contexts (identified by the `ch` prefix in Illumina probe identifiers) are excluded, as their interpretation differs from canonical CpG methylation and is outside the scope of this analysis [**Zhou2017**, **Pidsley2016**].

Sex-chromosome probes. Probes located on chromosomes X and Y are removed to prevent sex-driven confounding effects. Sex-chromosome CpGs exhibit substantially different methylation profiles between males and females, and including them in joint normalisation has been shown to introduce artificial sex-related bias into autosomal probes [**Wang2022**]. Since the datasets analysed here contain samples from both sexes and sex-specific methylation is not the object of investigation, removal of X/Y probes represents a conservative and standard practice [35].

The filtering lists are applied as a union: a probe is removed if it appears in at least one of the curated resources. Platform-specific manifests (HumanMethylation450 v1.2 for GSE69914 [**Illumina450K**]; MethylationEPIC v1.0 B5 for GSE225845 and GSE287331 [**IlluminaEPIC**]) are used for manifest-based validation, chromosome annotation, and non-CpG probe identification. The number of probes flagged by each resource, and the redundancy of flags across resources, are reported per-dataset in Section 4.3.

4.2.4 Statistical Transformation: β-to-M Values

The final preprocessing step transforms the filtered β-value matrix into M-values prior to statistical modelling. As described in Section **??**, β-values are bounded proportions in [0,1], defined as:

$$\beta = \frac{\max(y^{(M)}, 0)}{\max(y^{(M)}, 0) + \max(y^{(U)}, 0) + \alpha}, \tag{4.1}$$

where $y^{(M)}$ and $y^{(U)}$ denote the methylated and unmethylated fluorescence intensities respectively, and α is a small offset (typically 100) that stabilises the denominator at low signal intensities. While β-values are intuitive and directly interpretable as fractional methylation, their bounded support induces strong mean-dependent heteroscedasticity: variance is inflated near intermediate values ($\beta \approx 0.5$) and compressed near the boundaries, violating the homoscedasticity assumption required by standard linear models and empirical Bayes procedures [6].

The logit transformation to M-values,

$$M = \log_2\left(\frac{\beta + \varepsilon}{1 - \beta + \varepsilon}\right), \quad \varepsilon = 10^{-6}, \tag{4.2}$$

maps $\beta \in (0,1)$ to an unbounded real-valued scale and substantially stabilises the mean–variance relationship, as demonstrated theoretically and empirically by Du et al. [6]. The small offset ε is introduced to handle boundary values numerically. The resulting M-value distributions are approximately symmetric and exhibit markedly flatter variance profiles across the full methylation range, making them the standard input for differential methylation analysis with linear models, ComBat batch correction, and related inferential procedures [31, 32, 36].

It is important to note that M-values are adopted exclusively for statistical modelling: β-values are retained in parallel for biological interpretation, effect-size quantification, and visualisation, following the dual-representation convention recommended by Du et al. [6] and widely adopted in the literature [37, 36].

The transformation is applied to the complete post-filtering matrix for each dataset. Distributional diagnostics confirming the expected variance-stabilising effect are reported in Section 4.3.

4.3 Dataset-Specific Implementation

Data pre-processing represents a foundational stage in DNA methylation analysis, particularly in cancer epigenomics, where technical artefacts, probe-design biases, and unwanted sources of variation can obscure or distort genuine biological signals. Before performing feature selection, or machine-learning models, it is essential to ensure that the methylation matrices used for statistical modelling are accurate, and free from unreliable probes or batch-associated effects.

In the context of breast cancer, where early epigenetic alterations may be subtle and highly localized, careful pre-processing is crucial to avoid confounding Normal–Adjacent differences with technical noise. Several well-established sources of unreliability—cross-reactive probes, SNP-affected loci, non-CpG targets, Infinium Type I/II probe-design bias, and batch effects—must be addressed through a systematic and reproducible workflow.

The primary goal of this section is therefore to construct, for each dataset, a high-quality and biologically coherent methylation matrix suitable for downstream analyses. To this end, all datasets undergo the same structured pipeline: initial alignment and quality checks; removal of unreliable CpGs through literature-based technical filtering; and conversion from β-values to M-values for statistical modelling.

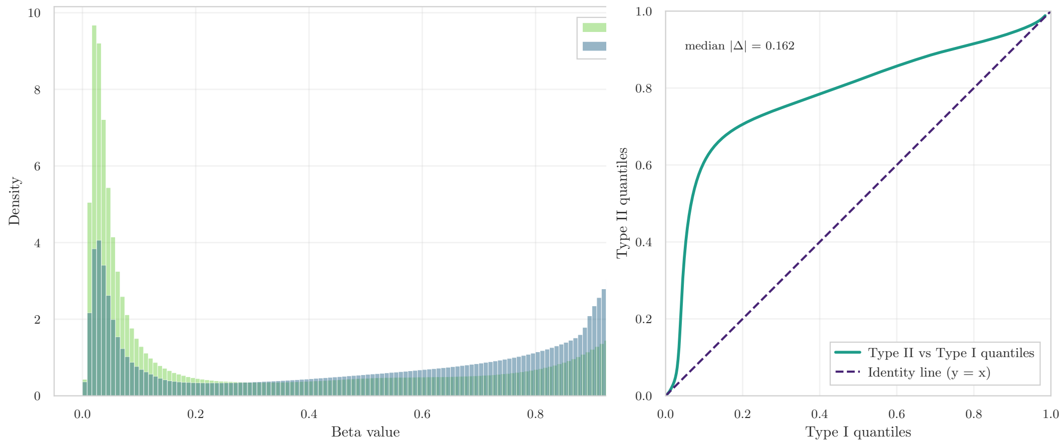
These steps ensure methodological consistency across the three cohorts (GSE69914 [4.3.1], GSE225845 [4.3.2], GSE287331 [4.3.3]) and provide a unified, denoised, and biologically meaningful set of features to be used in the subsequent analyses of early epigenetic alterations.

4.3.1 Dataset GSE69914

Step 0 – Initial Pre-processing The initial pre-processing stage verifies the structural integrity of the dataset. The β-value matrix and its corresponding phenotype table were imported from their Parquet representations. Both tables contained 397 samples, and the β-matrix included 485,512 CpGs, confirming full dataset availability at load time.

Sample identifiers in the phenotype table were inspected to ensure uniqueness and correct metadata structure. No duplicated `id_tissue` entries were detected (0 duplicates), and all required fields (`id_tissue`, `label`) were present (0 missing fields), indicating that the phenotype annotation is complete and coherent. Alignment between the phenotype table and the β-matrix was assessed by intersecting sample IDs. All samples were shared across matrices, yielding perfect correspondence (397/397 matched).

A probe-level and sample-level missingness check was then performed to verify the completeness of the methylation matrix. The entire dataset was found to be



(a) β-value distribution by Infinium probe design (Type I vs. Type II). The near-overlapping shapes indicate effective correction of probe-type bias.

(b) Q–Q plot comparing quantiles of Type II vs. Type I probes. The alignment to the diagonal ($y = x$) confirms balanced signal distributions after normalization.

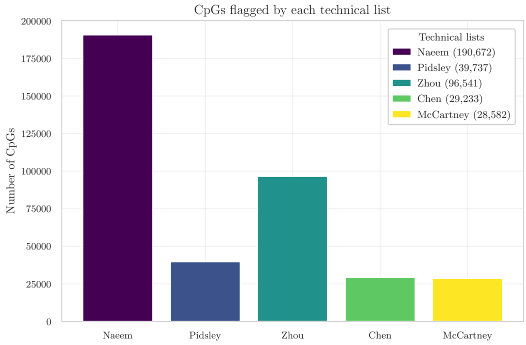
Figure 4.1: Diagnostic evaluation of Infinium Type I/II probe bias correction in the GSE69914 dataset. The consistent β-value and quantile distributions demonstrate the effectiveness of the BMIQ normalization previously applied.

fully observed, with 0 missing CpG values and 0 missing sample entries. This confirms that no early NaN filtering or imputation is required for this dataset.

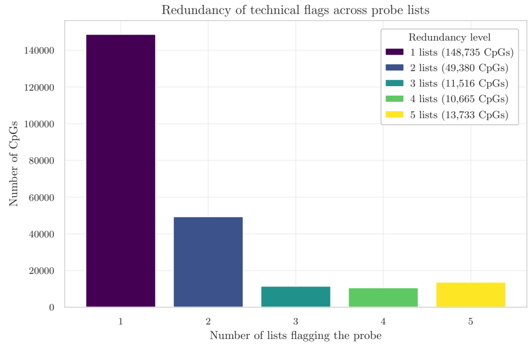
Step 1 – Infinium Type I/II Bias Diagnostics Raw methylation intensity data (*IDAT* files) were processed by the original authors using the `minfi` pipeline and `BMIQ` normalization, as reported in [3]. A well-known characteristic of Illumina Infinium microarrays is the **systematic technical bias between Type I and Type II probes**. This bias has been extensively documented in the literature, most notably by Maksimovic *et al.* [35], who showed that the two probe chemistries produce inherently different β-value distributions and therefore require appropriate probe-type normalization. Such differences, if uncorrected, can propagate into downstream statistical analyses.

To independently verify that this bias had been successfully mitigated in the present dataset, β-values were stratified by probe design (Type I vs. Type II) using the official Illumina 450K manifest from Bioconductor [38]. Following the diagnostic procedure of Teschendorff *et al.* [7], two complementary assessments were performed:

1. **Density comparison.** As shown in Figure 4.1a, the β-value distributions of Type I and Type II probes substantially overlap, indicating the absence of the



(a) CpGs flagged by each technical filtering resource (Naeem, Pidsley, Zhou, Chen, McCartney).



(b) Redundancy distribution of probes flagged by 1–5 lists.

Figure 4.2: Summary of technical filtering. **(a)** Intersection between the dataset and each external probe-filtering resource, showing the number of CpGs flagged by Naeem, Pidsley, Zhou, Chen, and McCartney lists. **(b)** Redundancy analysis indicating how many probes were simultaneously flagged by 1 to 5 independent resources, quantifying the concordance among technical annotations.

multi-modal distortion typical of unnormalized data.

2. **Quantile–quantile alignment.** The Q–Q plot in Figure 4.1b exhibits quite a diagonal trend for Type II vs. Type I quantiles, which is the expected signature of balanced probe-type scaling after normalization.

The resulting pattern closely matches the characteristic behaviour described by Teschendorff *et al.* [7] and Wang *et al.* [39], while being fully consistent with the probe-type discrepancies originally analysed by Maksimovic *et al.* [35]. Together, these diagnostics confirm that the dataset has undergone effective Type I/II probe bias correction.

Step 2 – Technical Filtering Technical filtering aims to remove unreliable or biologically confounded probes prior to statistical modelling. This stage reduces noise, improves downstream reproducibility, and ensures that only high-confidence CpG loci are retained for analysis (Figure 4.2).

Exclusion of technical probe sets. A collection of curated probe resources was used to exclude loci known to exhibit technical artefacts (see Appendix **??** for details). These include probes affected by common genetic variants at the CpG or single-base extension site [40, 19], probes with demonstrated multi-mapping or off-target hybridisation [41, 42], and probes flagged by the `MASK_*` reliability annotations [40]. Additional quality control criteria follow the hierarchical filtering strategy of Naeem et al. [43], which targets multi-mapping loci, repeat structures, and disruptive SNPs.

The overlaps between these resources and the β-matrix were substantial (Figure 4.2a). A total of 190,672 probes were flagged by Naeem et al., 39,737 by Pidsley et al., 29,233 by Chen et al., 96,541 by the Zhou `MASK_*` fields, and 28,582 by McCartney et al. These lists are known to partially overlap—particularly those focused on multi-mapping and SNP-associated unreliability—and their union resulted in the removal of 225,426 unique CpGs. After this operation, 260,086 CpGs remained.

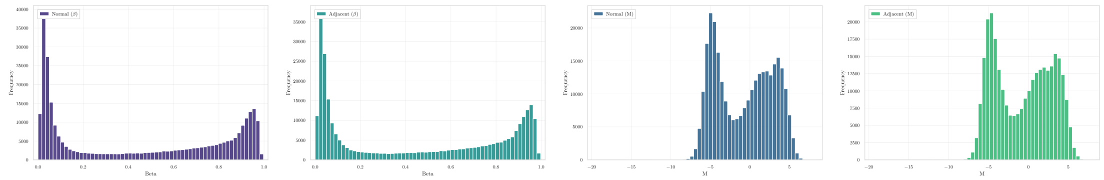
Annotation-based filtering (Illumina manifest). To ensure consistency with the manufacturer's official annotation, the HumanMethylation450 v1.2 manifest [44] was used to validate probe identity and genomic mapping. Through this annotation, probes lacking valid chromosome information, those with undefined genomic coordinates, and those targeting non-CpG contexts (IDs beginning with "`ch`") were identified and removed. This step aligns the working feature set with Illumina's reference genome definition and follows the recommendations of Zhou et al. [40] and Pidsley et al. [19]. A total of 875 non-CpG probes were excluded, reducing the feature space to 259,211 CpGs.

Removal of sex-chromosome probes (X/Y). To avoid sex-related confounding effects and to maintain cross-dataset consistency, all probes located on chromosomes X and Y were removed based on manifest annotations. This choice is consistent with established preprocessing workflows for Illumina 450K data, where probes on sex chromosomes are excluded when male and female samples are mixed, as recommended by Maksimovic et al. [35]. Moreover, sex-chromosome CpGs exhibit substantially different methylation profiles between males and females, and joint normalization of autosomal and sex-chromosome probes has been shown to introduce artificial sex-related bias into thousands of autosomal CpGs [45]. Removing X/Y probes therefore provides a conservative and robust strategy when sex-specific methylation is not under investigation. This filtering step eliminated 7,728 CpGs, producing a final autosomal working set of 251,483 CpGs.

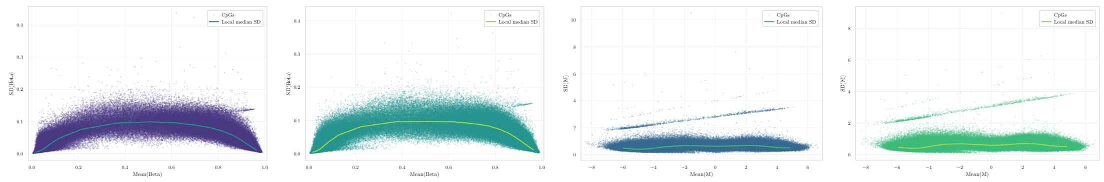
Overall, the initial set of 485,512 CpGs was reduced to 251,483 CpGs after applying all technical lists, manifest-based validation, and sex-chromosome exclusion. In total, 234,029 unique CpGs were removed across all criteria. A per-probe summary file `removed_cpgs_all_filters_summary_gse69914.csv` records, for each excluded probe, which resources contributed to its removal, ensuring full reproducibility of the filtering pipeline.

Table 4.1: CpG filtering summary for the GSE69914 dataset.

Initial CpGs	Naeem	Pidsley	Zhou	Chen	McCartney	Non-CpG	X/Y	Final Cp
485,512	190,672	39,737	96,541	29,233	28,582	875	7,728	251,483



(a) Normal (β) **(b)** Adjacent (β) **(c)** Normal (M) **(d)** Adjacent (M)



(e) Mean–SD Normal (β) **(f)** Mean–SD Adjacent (β) **(g)** Mean–SD Normal (M) **(h)** Mean–SD Adjacent (M)

Figure 4.3: Distributional comparison between β and M quantifications in Normal and Normal-adjacent tissues. Top row: β exhibits a bounded, U-shaped distribution, whereas M shows an approximately symmetric profile. Bottom row: The substantial mean-dependent variance in β is dramatically reduced after transformation, confirming that M-values provide a more homoscedastic scale for statistical modeling.

Step 3 – β to M Conversion Illumina Infinium methylation arrays measure methylated ($y^{(M)}$) and unmethylated ($y^{(U)}$) signal intensities per CpG. The processed dataset provides only β-values,

$$\beta = \frac{y^{(M)}}{y^{(M)} + y^{(U)}},$$

a bounded proportion in [0,1] characterised by strong mean–variance dependence. Following the foundational analysis of Du *et al.* [6], all β-values were transformed into M-values using the logit relation

$$M = \log_2\left(\frac{\beta + \varepsilon}{1 - \beta + \varepsilon}\right), \qquad \varepsilon = 10^{-6},$$

which yields an approximately homoscedastic scale more suitable for linear modelling, empirical Bayes moderation, ComBat batch correction, and regression-based inference.

Importantly, the adoption of M-values is not only theoretically motivated but also widely implemented in practice. Several high-impact studies routinely perform statistical modelling in M-space while retaining β-values solely for biological interpretation and effect-size visualisation. Examples include the differential methylation analysis of purified human blood cell types by Reinius *et al.* [37], the methodological framework underlying the `minfi` pipeline described by Triche *et al.* [31], and oncological applications such as the tumour hypoxia study by Thienpont *et al.* [36], where all β-values were explicitly transformed to M-values prior to statistical testing. These examples illustrate that M-values have become a de facto standard for inferential analyses in Illumina 450K/EPIC workflows.

The conversion was applied to the complete post-filtering matrix (251,483 CpGs across 397 samples).

Distributional consequences of the transformation. Figure 4.3 illustrates in detail how the $\beta \to M$ mapping reshapes the empirical distribution of methylation. Normal and Adjacent tissues display the expected U-shaped profiles in β-space, reflecting biological accumulation near fully unmethylated or fully methylated states. After logit transformation, the corresponding M-value distributions become substantially more symmetric and comparable across groups, indicating that the transformation regularises the scale without distorting biological structure.

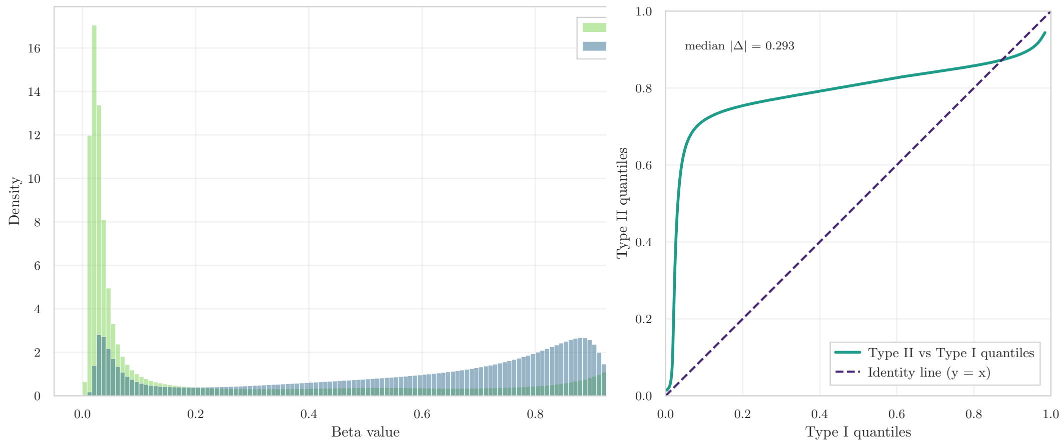
The mean–SD relationships further highlight this effect. In β-space, variance is strongly mean-dependent and inflates near the boundaries, whereas M-values exhibit markedly flatter SD profiles. This variance-stabilising behaviour is consistent with the theoretical findings of Du *et al.* [6] and with the practical motivations underlying the adoption of M-values in modern methylation studies.

4.3.2 Dataset GSE225845

Step 0 – Initial Pre-processing The β-value matrix was successfully imported from its Parquet representation, yielding 477 samples and 750,426 CpGs, reflecting the high dimensionality of the Illumina EPIC platform. The corresponding phenotype table contained the same number of samples (477 entries) and provided all required metadata fields with no inconsistencies.

Sample identifiers were inspected to ensure uniqueness and correct formatting. No duplicated `id_tissue` values were detected, and all metadata columns required for downstream analysis were present (0 missing fields). Alignment between phenotype records and the methylation matrix was confirmed by intersecting sample IDs, producing perfect correspondence (477/477 matched).

A probe-level and sample-level missingness check verified the completeness of the methylation data. The dataset was fully observed, with 0 missing CpG values and 0 missing sample entries. Accordingly, no early NaN filtering or imputation was required before proceeding to probe-type diagnostics and technical filtering.



(a) β-value distributions for Infinium Type I and Type II probes. The two probe families exhibit markedly different empirical shapes, indicating residual probe-design bias.

(b) Q–Q comparison of Type II vs. Type I β-value quantiles. The substantial deviation from the identity line ($y = x$), with median absolute deviation 0.2933, indicates incomplete probe-type alignment.

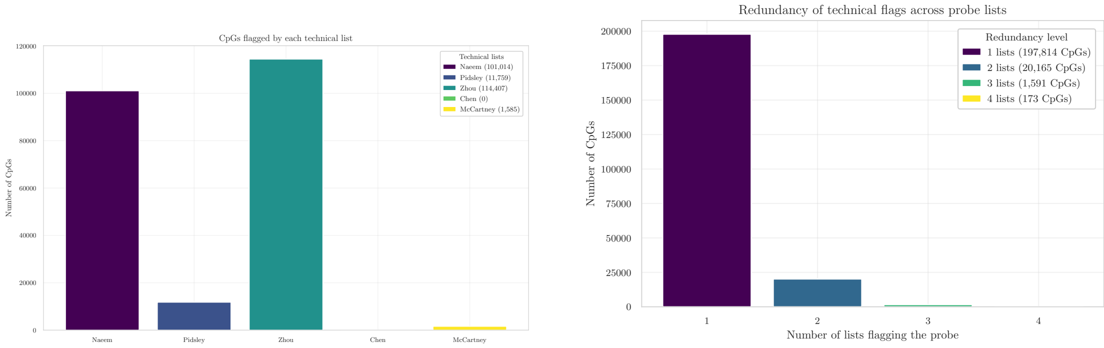
Figure 4.4: Diagnostic evaluation of Infinium Type I/II probe-design bias. Both the density distributions and the Q–Q pattern reveal that the two probe chemistries retain systematically different scaling behaviour, indicating that residual Type I/II bias persists in the dataset.

Step 1 – Infinium Type I/II Bias Diagnostics All probes were stratified, this yielded 119,760 Type I and 627,842 Type II probes.

The density representation of the two probe families (Figure 4.4a) reveals a marked discrepancy between their empirical β-value distributions. Type I probes exhibit a highly concentrated pattern near the unmethylated and fully methylated boundaries, whereas Type II probes display a broader and right-shifted distribution. Such divergence indicates that the two probe chemistries retain systematically different scaling behaviour.

This finding is reinforced by the quantile–quantile comparison in Figure 4.4b. Type II quantiles deviate substantially from the identity line, with a median absolute deviation of 0.2933, demonstrating that the expected alignment between the two probe types is not present. The curvature of the Q–Q profile confirms the presence of a residual probe-design imbalance that has not been fully corrected by preprocessing.

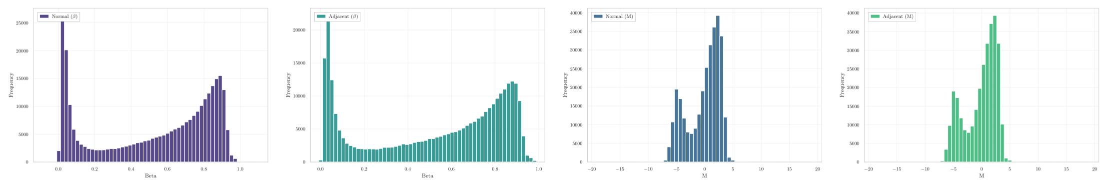
Step 2 – Technical Filtering Beginning from an initial set of 750,426 CpGs, the intersection with the external probe-filter lists showed substantial overlap across multiple sources.



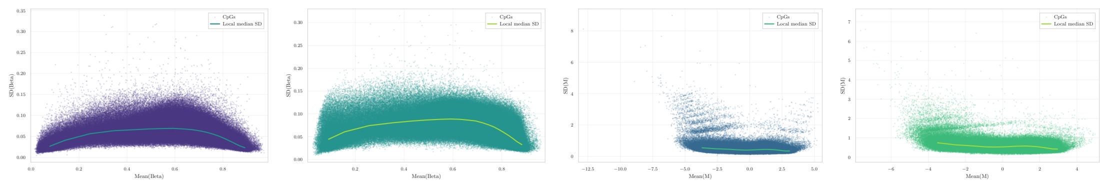
(a) Number of CpGs flagged by each technical list.

(b) Redundancy distribution: CpGs flagged by 1–4 lists.

Figure 4.5: Summary of technical filtering for the methylation dataset. **(a)** Overlap between CpGs and the Naeem, Pidsley, Zhou, Chen, and McCartney lists. **(b)** Redundancy analysis quantifying concordance among the technical annotations.



(a) Normal (β) **(b)** Adjacent (β) **(c)** Normal (M) **(d)** Adjacent (M)



(e) Mean–SD Normal (β) **(f)** Mean–SD Adjacent (β) **(g)** Mean–SD Normal (M) **(h)** Mean–SD Adjacent (M)

Figure 4.6: Distributional and variance-structure comparison of β and M values in Normal and Adjacent breast tissue. Top row: β- and M-value histograms, showing the characteristic U-shape in β-space and the approximate symmetry induced by the logit transformation. Bottom row: mean–SD relationships illustrating how the strong mean-dependent variance in β-space is markedly stabilised in M-space.

Exclusion of technical probe sets. The curated technical resources flagged a substantial fraction of probes as potentially unreliable (Figure 4.2a): 101,014 CpGs were flagged by Naeem, 11,759 by Pidsley, 114,407 by the Zhou fields, 0 by Chen,

Table 4.2: CpG filtering summary for the GSE225845 dataset.

Initial CpGs	Naeem	Pidsley	Zhou	ChenX	McCartney	Non-CpG	X/Y	Final C
750,426	101,014	11,759	114,407	0	1,585	773	14,071	530,6

and 1,585 by McCartney. The union of all technical lists resulted in the removal of 204,899 unique CpGs, leaving 545,527 CpGs for subsequent steps.

Annotation-based filtering (Illumina manifest). Using the EPIC v1.0 manifest, all probes lacking valid genomic coordinates or targeting non-CpG contexts were removed. This step eliminated 773 non-CpG loci, reducing the working set to 544,754 CpGs.

Removal of sex-chromosome probes (X/Y). Probes mapped to chromosomes X and Y were excluded. A total of 14,071 CpGs were removed, producing a filtered autosomal set of 530,683 CpGs.

Across all steps, 219,743 unique CpGs were removed from the initial 750,426, yielding a final feature set of 530,683 CpGs. A detailed per-probe summary of the filtering decisions is provided in `removed_cpgs_all_filters_summary_gse225845.csv`.

Step 3 – β to M Conversion All β-values were transformed into M-values using the standard logit relation yielding an unbounded and approximately homoscedastic scale suitable for linear modelling and batch correction procedures. The transformed dataset, denoted `M_clean`, preserves the same sample structure and CpG ordering as the β matrix and was saved in Parquet format for downstream analysis.

Distributional effects of the transformation. The empirical distributions of Normal and Adjacent samples exhibit the characteristic U-shape in β-space and the expected approximately symmetric profiles after conversion. The mean–SD relationship demonstrates the well-known variance inflation at intermediate β levels, which becomes substantially stabilised in M-space (Figure 4.6). These patterns confirm that the M-value representation provides a more appropriate scale for statistical inference.

The filtered β-matrix (530,683 CpGs, 477 samples) was successfully converted into the corresponding M-value representation and stored as a Parquet object.

4.3.3 Dataset GSE287331

Missingness filtering and imputation I quantified the amount and structure of missing values in the β-matrix. Probe-wise missingness is a well-known source of unreliability in Illumina methylation arrays, and several studies recommend removing CpGs that exhibit low measurement quality or an excessive proportion of missing values. For example, Islam *et al.* [46] excluded probes with missing β-values in more than 2% of samples in their pediatric tissue methylation analysis, while Mansell *et al.* [47] applied a similar 1% threshold for detection failure in their EPIC-array quality control pipeline.

Guided by this evidence and the large number of CpG sites available, I adopted a stringent probe-wise filtering criterion and removed all CpGs missing in more than 1% of samples. This choice prioritises high-confidence measurements while minimising reliance on imputed values.

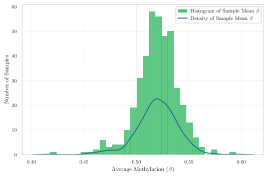
After filtering, only a negligible fraction of values remained missing in the working matrix (58 561 NaN, corresponding to 0.0187% of all entries). Given this extremely low level of sparsity, median-based imputation is both statistically robust and distribution-preserving. Following current recommendations, where missing methylation values are replaced by probe-wise medians to avoid introducing group-specific bias, all residual NaN were imputed using the median β-value of each CpG across samples. This approach is consistent with recent harmonization methods such as mLiftOver, where missing methylation values are explicitly replaced using probe-wise median substitution to preserve biologically plausible levels [48].

Evidence of Missing Infinium Type I/II Probe Bias Correction During the exploratory analysis performed uniformly across all three datasets, the per-sample mean β-value distributions (Fig. 4.7a, Fig. 4.7b, Fig. 4.7c) revealed a marked inconsistency. Whereas `GSE69914` and `GSE225845` displayed a single broad peak of global methylation levels, `GSE287331` showed a much narrower distribution, shifted towards higher β-values and characterized by two distinct modes in the high-β range.

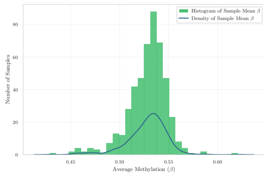
This unusual right-shifted bimodality—largely absent in the other datasets—strongly indicated that a key preprocessing step, namely Infinium Type I/II probe-design bias correction, had not been applied in the processed data released by the authors.

To confirm this hypothesis, I stratified all probes by Infinium design type (Type I vs. Type II) using the official Illumina EPIC v1.0 manifest, obtained from the corresponding Bioconductor annotation package [38]. Using this manifest, the β-value distributions were separated by probe design and inspected through a dedicated diagnostic plot (Fig. 4.8). According to the expected behaviour described by Teschendorff et al. [7], datasets that have undergone BMIQ normalization should display nearly overlapping β-value distributions for the two probe types.

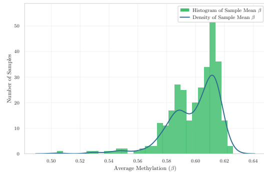
In contrast, the Q–Q plot comparing Type I and Type II probes revealed a markedly elevated median absolute deviation (median $|\Delta| = 0.380$), indicating a strong and systematic quantile-wise shift between the two designs. Such a deviation is fully incompatible with BMIQ-corrected data, which should instead exhibit deviations close to zero across all quantiles. After applying BMIQ, this deviation



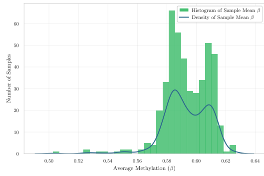
(a) GSE69914



(b) GSE225845



(c) GSE287331 pre BMIQ



(d) GSE287331 post BMIQ

Figure 4.7: Per-sample mean β distributions for the three datasets. `GSE69914` and `GSE225845` show a single broad peak, whereas `GSE287331` displays a right-shifted bimodal pattern. After BMIQ, the two peaks become slightly closer and more similar in height, but the bimodality persists.

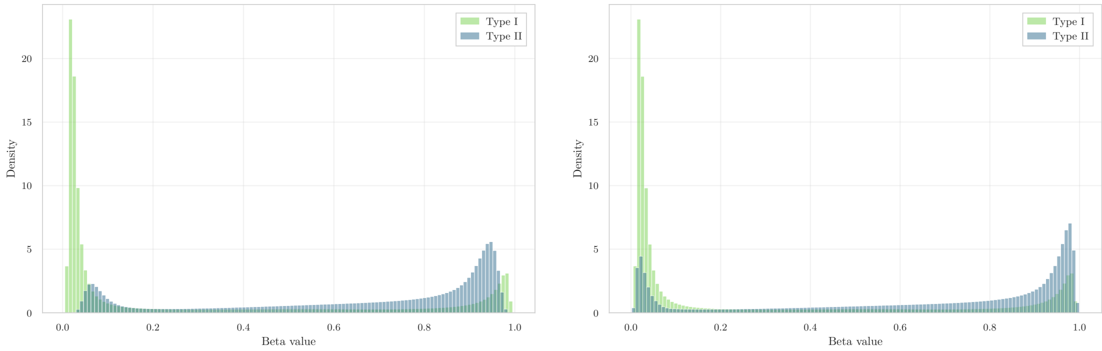
decreases only marginally ($|\Delta| : 0.380 \to 0.370$), confirming that the correction improves probe-type alignment but only to a limited extent.

This modest improvement is also clearly visible in the diagnostic density plots (Fig. 4.8b). After BMIQ, the rightmost portion of the distribution (near $\beta \approx 1$) shows Type II probes becoming visibly more similar to Type I, and the strong asymmetry between the two curves is partially reduced. Nevertheless, the overall shapes of the two probe distributions remain distinct, and the characteristic "double-tail" pattern of the uncorrected dataset is still present after normalization. BMIQ therefore warps the distribution in the expected direction, but the magnitude of the adjustment is small and insufficient to reconcile the two probe types to the degree observed in datasets processed with a full bias-correction pipeline.

A similar behaviour can be observed in the per-sample mean β distributions: after BMIQ, the two peaks become slightly higher and closer to one another (Fig. 4.7d), indicating that the algorithm partially reduces the gap between the two dominant modes. However, even after correction, the distribution remains visibly and consistently *bimodal*, suggesting that BMIQ mitigates only weakly the underlying discrepancy and that sample-level heterogeneity continues to dominate the global methylation structure.

Taken together, these observations provide clear visual evidence that no probe-design bias correction was applied in the original processing. To further support this conclusion, I examined the official publication [24]. The paper carefully describes the acquisition, processing, and analytical use of Illumina EPIC β-values, but it does *not* mention BMIQ, SWAN, noob, or any other method aimed at correcting Infinium probe-type bias. The absence of such a statement—combined with the strong visual mismatch between Type I and Type II probe distributions—confirms that no probe-design bias correction was applied before the release of the dataset.

For these reasons, applying a dedicated Infinium Type I/II bias correction step is necessary before performing any comparative analysis.



(a) GSE287331 pre BMIQ **(b)** GSE287331 post BMIQ

Figure 4.8: Type I vs. Type II probe design comparison before and after BMIQ. After correction, Type II probes become more similar to Type I in the high-β region (especially near $\beta \approx 1$), and the overall discrepancy is reduced ($|\Delta| : 0.380 \rightarrow 0.370$), though probe types remain clearly distinct.

BMIQ correction: implementation and methodological considerations
Given the clear evidence of missing Infinium Type I/II probe-bias correction, I applied the canonical BMIQ algorithm [7], using the implementation provided by the `wateRmelon` package [49], to harmonize the distributions of Type I and Type II probes for each sample. To ensure full reproducibility and strict memory control on the extremely wide EPIC matrix, I implemented a *sequential, sample-wise BMIQ correction*, avoiding any between-sample pooling or multi-core parallelization. Each sample's β vector was processed independently using a safe wrapper around `wateRmelon::BMIQ`, which first verified the availability of a sufficient number of non-missing Type I and Type II probes and then returned the corrected β-values. The corrected matrix was updated in place, preserving the schema `id_tissue | CpGs... | label` and subsequently written to a single Parquet file. This design guarantees that no cross-sample information influenced the correction and that the warping applied by BMIQ reflects exclusively the distributional characteristics of each individual sample.

While BMIQ is widely used to mitigate probe-design bias, multiple studies have raised important methodological caveats regarding normalization procedures that operate on the full distribution of methylation values. Yousefi *et al.* [50] highlighted that quantile-based adjustments and related normalization strategies may "come at the cost of *artificially reduced biological variability*," a phenomenon that is particularly undesirable when studying biological groups expected to differ globally in their methylation profiles. Similarly, Liu *et al.* [51] reported that, when methylation alterations are extensive in size and number, certain normalization methods can "*remove biological signal*," effectively shrinking meaningful differences between groups.

More recently, Welsh *et al.* [52] performed a systematic evaluation of normalization methods for EPIC arrays and showed that approaches enforcing strong distributional alignment may distort real biological structure. In particular, the authors observed that quantile-based normalization "*forces the empirical marginal distributions of the samples to be the same, removing all variation in this statistic*," and reported that BMIQ, in some settings, yielded "*distorted and discontinuous*" distributions. Such findings provide a conceptual explanation for the behaviour observed: although BMIQ was applied strictly per-sample, its within-array warping can attenuate global differences between Normal, Adjacent, and Tumor samples when these differences are manifested as broad shifts in the underlying β-value distribution.

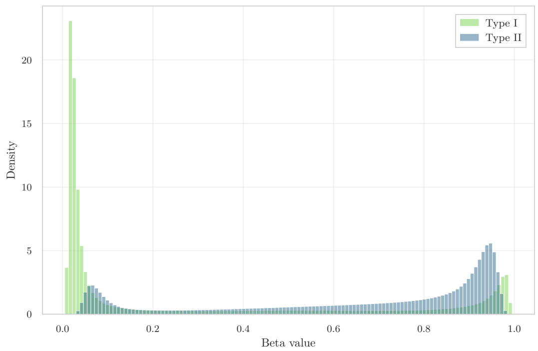
Together, these considerations emphasise both the necessity and the limitations of probe-design bias correction. Applying BMIQ was essential to remove the technical Type I/II bias absent in the original processing, yet its known tendency to reduce distributional heterogeneity across groups must be carefully taken into account when interpreting group-level contrasts in downstream analyses.

Importantly, the modest degree of probe-type harmonization is accompanied by a noticeable flattening of biological structure: as will be shown in the low-dimensional embedding analysis (Section 3.3.3), PCA, UMAP, and t-SNE projections reveal that the Normal, Adjacent, and Tumor groups become *less clearly separable* after BMIQ. For this reason, all downstream analyses are carried out on the **non-BMIQ version** of the dataset, which preserves sharper cluster boundaries and better reflects the underlying biological signal.

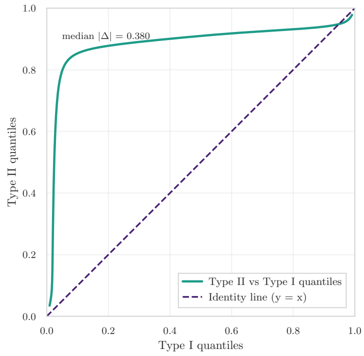
This strong separation is also quantitatively confirmed by silhouette analysis: the average silhouette coefficient *before* BMIQ was $\text{sil}_{\text{pre}} = 0.2876$, but *after* BMIQ correction it declined to $\text{sil}_{\text{post}} = 0.2537$. Thus, BMIQ *reduced*–even if not significantly–the natural clusterability of the data, collapsing genuine biological differences—particularly the clear separation between Normal and Adjacent samples.

For this reason, and in accordance with the principle of preserving authentic biological signal, the BMIQ-normalised version was not adopted.

Step 0 – Initial Pre-processing The dataset was imported from its Parquet representation, yielding an initial β-value matrix of 446 samples and 866,552 CpGs. The accompanying phenotype table reported the same number of entries and contained all required metadata fields. Before continuing with probe-level preprocessing, the dataset underwent a dedicated missingness-cleaning stage to ensure high-quality input for downstream analysis. After harmonising phenotype labels and applying dataset-specific selection rules, only samples belonging to the three core tissue states of interest—Normal, Adjacent, and Tumour—were kept.



(a) Type I vs. Type II density (pre BMIQ).



(b) Q–Q comparison of Type II vs. Type I probes (pre BMIQ). The median absolute deviation of 0.380 indicates a strong quantile-wise shift, confirming the absence of probe-type normalization in the original dataset.

Figure 4.9: Diagnostic evaluation of Infinium probe–design bias before and after BMIQ. The correction reduces the discrepancy only marginally, and substantial distributional differences remain between probe types.

This resulted in a final working cohort of 311 samples, used for all downstream preprocessing and analysis.

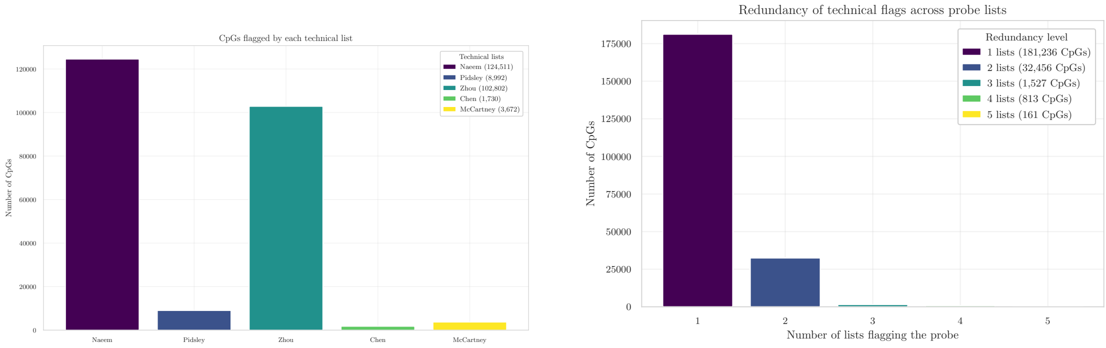
Missingness filtering and imputation. Probe-wise missingness was quantified across the β-matrix. Consistent with recommendations from Islam *et al.* [46] and Mansell *et al.* [47], a stringent threshold was applied: all CpG sites missing in more than 1% of samples were removed. This filter ensures high-confidence measurements while leveraging the large dimensionality of the EPIC platform.

After applying this criterion, the remaining level of sparsity was extremely low, with only 58,561 NaN values (0.0187% of all entries). These residual missing values were imputed using probe-wise median substitution, a strategy that avoids introducing group-specific bias and is also adopted in recent harmonisation workflows such as mLiftOver [48].

The resulting β-matrix contained no missing values.

Data integrity verification. After missingness filtering and imputation, the final working matrix consisted of 311 samples and 702,916 CpGs. All entries were fully observed (0 residual NaN).

Step 1 – Infinium Type I/II Bias Diagnostics Using the official EPIC v1.0 manifest, 115,705 Type I and 585,514 Type II probes were identified. The β-value distributions of the two chemistries showed a marked mismatch (Figure 4.9a), with



(a) CpGs flagged by each technical list.

(b) Redundancy of probes flagged by 1–5 lists.

Figure 4.10: Summary of technical filtering for the methylation dataset. **(a)** Number of CpGs intersecting each curated resource. **(b)** Redundancy analysis capturing concordance across filtering lists.

Type II probes displaying a strong right-shifted profile. The quantile–quantile comparison confirmed a substantial systematic offset, with a median absolute deviation of 0.380 (Figure 4.9b), indicating the absence of any Infinium probe-design bias correction in the originally released data.

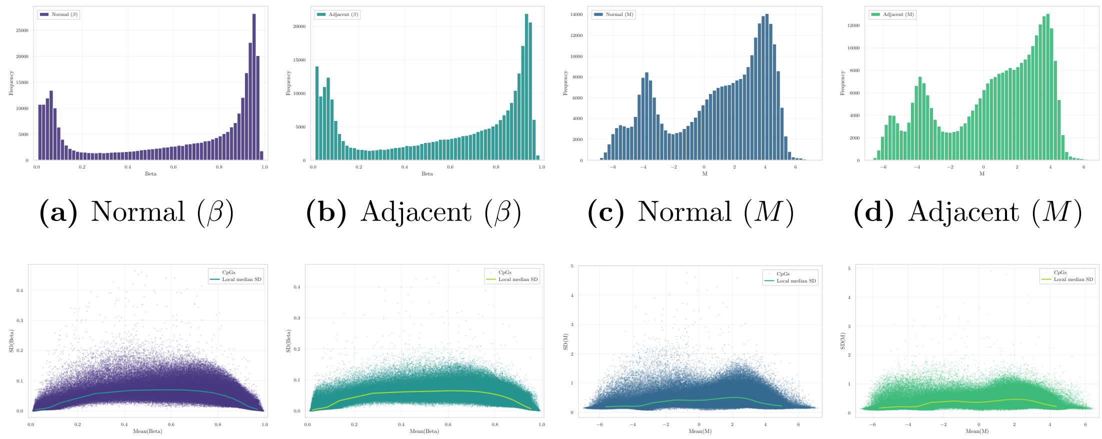
Given this discrepancy, a per-sample BMIQ normalization step was applied using the `wateRmelon` implementation, operating strictly on individual samples to minimise cross-sample distortion. Post-correction diagnostics showed a modest reduction in probe-type mismatch ($|\Delta|$: 0.380 $\rightarrow$ 0.370) and a slight improvement in the alignment of high-β regions, but the overall distributional discrepancy remained substantial. Due to this limited harmonization and the known tendency of BMIQ to attenuate biological differences when global distributional shifts are present, the corrected matrix is not used for downstream modelling. All subsequent analyses therefore rely on the non-BMIQ version of the dataset, which preserves clearer group-level structure and avoids normalization-induced shrinkage (for more details, see *01 – Report Data Exploration & Visualization.*).

Step 2 – Technical Filtering Technical probe filtering excluded CpG loci affected by unreliable hybridisation, SNP interference, multi-mapping or annotation inconsistencies. Across the established probe-filtering resources, 124,511 CpGs were flagged by Naeem, 8,992 by the Pidsley blacklist, 102,802 by Zhou's fields, 1,730 by Chen's cross-reactive probe set, and 3,672 by McCartney's cross-hybridisation catalogue. The union of these lists removed 203,114 unique CpGs, leaving 499,804 CpGs prior to annotation-based filtering.

Annotation-based filtering (Illumina manifest). Using the EPIC v1.0

Table 4.3: CpG filtering summary for the GSE287331 dataset.

Initial CpGs	Naeem	Pidsley	Zhou	Chen	McCartney	Non-CpG	X/Y	Final Cp
866,552	124,511	8,992	102,802	1,730	3,672	500	12,579	486,72



(a) Normal (β) **(b)** Adjacent (β) **(c)** Normal (M) **(d)** Adjacent (M)

(e) Mean–SD (Normal, β) **(f)** Mean–SD (Adjacent, β) **(g)** Mean–SD (Normal, M) **(h)** Mean–SD (Adjacent, M)

Figure 4.11: Distributional and variance-structure comparison of β and M values in Normal and Adjacent breast tissue. Top row: empirical β- and M-value distributions, illustrating the U-shaped behaviour of β-values and the symmetry introduced by the logit transformation. Bottom row: mean–SD relationships showing strong mean-dependent variance in β-space and its stabilisation in M-space, supporting the use of M-values for downstream modelling.

manifest, probes lacking valid genomic coordinates or targeting non-CpG contexts were removed. This eliminated 500 non-CpG probes, reducing the feature set to 499,304 CpGs.

Removal of sex-chromosome probes (X/Y). Probes mapped to chromosomes X and Y were removed to avoid sex-driven confounding. A total of 12,579 CpGs were discarded, resulting in an autosomal working set of 486,725 CpGs.

Across all filters, 216,193 unique CpGs were removed from the initial 866,552, leaving a final autosomal set of 486,725 CpGs. A probe-level removal summary is provided in `removed_cpgs_all_filters_summary_gse287331.csv`.

Step 3 – β to M Conversion Given the strong mean–variance dependence intrinsic to β-values, the working matrix obtained after technical and invariance filtering (486,725 CpGs across 311 samples) was transformed into M-values. This transformation yields an approximately homoscedastic scale and is the standard

Table 4.4: Summary of sample size and CpG dimensionality before and after pre-processing.

Dataset	Platform	Normal	Adjacent	CpGs *before* pre-processing	CpGs *after* pre-pr
GSE69914	450K	50	42	485,512	251,483
GSE225845	EPIC	113	140	750,426	530,683
GSE287331	EPIC	182	60	866,554	486,725

input for linear models, empirical Bayes moderation, and batch-effect correction.

The transformation was applied to all CpG columns (284,745 CpGs, excluding `id_tissue` and `label`). The resulting object, `M_clean`, preserves the same schema as the β-matrix and was stored as a memory-efficient `LazyFrame`.

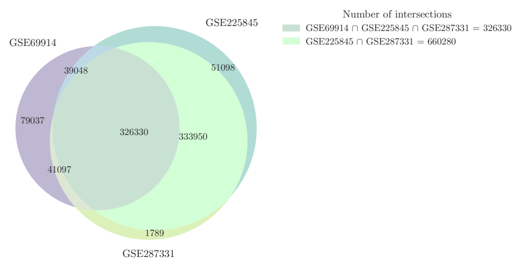
Distributional effects of the $\beta \to M$ transformation. Figure 4.11 illustrates the impact of the transformation on the empirical data distribution. As expected, the U-shaped β-profiles for Normal and Adjacent tissues become substantially more symmetric after conversion, while the pronounced mean–SD dependence flattens into a markedly more stable variance structure across the full range of M-values. These properties make M-values preferable for statistical modelling, whereas β-values are retained for biological interpretation and visualisation.

4.4 Cross-Dataset Effects of Preprocessing

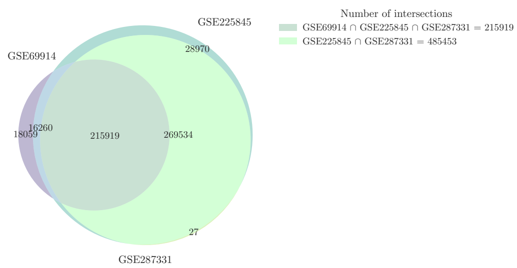
After completing the intra-dataset pre-processing for each cohort, a concise inter-dataset analysis is performed to summarise the effects of the pipeline and to motivate the subsequent feature selection step.

Pverview of the pre-processing pipeline All three datasets underwent a harmonised pre-processing workflow, including initial integrity checks, literature-based technical filtering, manifest-driven probe validation, removal of sex-chromosome loci, and transformation from β-values to M-values. While the specific implementation details differ slightly depending on platform characteristics and data quality, the conceptual structure of the pipeline is consistent across datasets.

Table 4.4 reports, for each dataset, the number of samples retained and the dimensionality of the methylation matrix before and after pre-processing. Despite substantial differences in initial platform coverage (450K vs EPIC), technical filtering consistently removes a large fraction of probes, yielding cleaner and more reliable feature spaces.



(a) CpG overlap across datasets before pre-processing.



(b) CpG overlap across datasets after pre-processing.

Figure 4.12: Comparison of CpG overlap across datasets before and after pre-processing.

CpG overlap across datasets To further characterise the relationship between datasets, CpG overlap was evaluated both before and after pre-processing. Figure 4.12a illustrates the overlap of CpG identifiers across the three datasets prior to filtering, reflecting platform design constraints and baseline annotation differences. As expected, the intersection is dominated by probes shared between the two EPIC datasets, with a smaller but substantial core common to all three cohorts.

Figure 4.12b reports the corresponding overlap after the complete pre-processing pipeline. Although the total number of CpGs is markedly reduced, a large shared core remains, indicating that technical filtering primarily removes platform- or annotation-specific probes while preserving a common, high-confidence CpG backbone.

4.5 Post-Preprocessing Data Structure and Feature Selection Rationale

Despite extensive technical filtering, the post-processed datasets remain highly dimensional, with hundreds of thousands of CpGs retained in each cohort. Moreover, the large shared CpG intersection across datasets suggests that technical artefacts have been effectively mitigated, shifting the primary challenge from data quality to signal extraction.

At this stage, the limiting factor is no longer technical reliability but statistical and biological relevance. The remaining feature space is expected to contain a substantial proportion of CpGs that are either weakly informative or irrelevant for distinguishing Normal and Adjacent breast tissue. Consequently, a dedicated feature selection strategy is required to identify a reduced subset of CpGs that captures biologically meaningful variation while improving interpretability, robustness, and downstream modelling performance.

The next chapter therefore focuses on feature selection methodologies applied to these pre-processed datasets, building on the stable and harmonised CpG space established here.

Chapter 5

Feature selection

REPORT 03

Chapter 6

Model training and evaluation

REPORT 04

Chapter 7

Dynamic System for biomarker detection

REPORT 05

Chapter 8

Conclusion and future work

CONCLUSIONI E PROSPETTIVE DI FUTURO

Appendix A

Filtering lists

This appendix details the major technical filtering lists and annotations applied in Illumina 450K and EPIC methylation arrays to remove unreliable or ambiguous CpG probes.

A.0.1 Reducing the risk of false discovery enabling identification of biologically significant genome-wide methylation status using the humanmethylation450 array

Naeem *et al.* [43] proposed a structured filtering framework to reduce false discoveries by sequentially discarding probes based on hybridisation specificity and genomic integrity (Figure A.1). The main exclusion steps are:

- Multi-mapping or cross-hybridising probes ⇒ *discard.*

- Probes overlapping repetitive elements (LINE/SINE/ALU) ⇒ *discard.*

- Probes targeting regions with INDELs ⇒ *discard.*

- Probes overlapping SNPs:

    – SNP at CpG or extension site ⇒ *discard.*

    – SNP near target but not interfering in bisulfite space ⇒ *keep.*

The resulting "discard" set therefore removes probes affected by cross-reactivity, structural polymorphisms (INDELs), and SNPs that alter CpG interrogation.

A.0.2 Discovery of cross-reactive probes and polymorphic cpgs in the illumina infinium humanmethylation450 microarray

Chen *et al.* [41] empirically identified probes in the 450K array that exhibit off-target hybridisation or overlap with common SNPs. For this project, only the **cross-reactive list** is available and used. This list enumerates ∼29k multi-mapping probes whose methylation signal cannot be uniquely attributed to one genomic locus.

A.0.3 Critical evaluation of the Illumina MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling

Pidsley *et al.* [19] provide a platform-level assessment of the EPIC array, with explicit *annotated probe lists* that flag probes whose methylation signal may be confounded by design- or genome-related artefacts. The focus is on probe categories and positions where technical bias arises, rather than on a universal, prescriptive blacklist. In particular, they catalogue:

- **Cross-hybridising CpG-targeting probes:** CpG probes showing sequence homology (off-target matches) to additional genomic loci, yielding non-unique hybridisation and potentially inflated or ambiguous β signals.

- **Cross-hybridising non-CpG-targeting probes:** off-target issues among CNG/non-CpG probes; these are typically excluded in CpG-centric analyses due to limited interpretability and higher risk of artefacts.

- **Probes overlapping common genetic variation:** annotation of variants from population data at three critical positions:
  - *At the interrogated CpG* (polymorphic CpG) — directly affects the presence of the CpG dinucleotide and the measured methylation state;
  - *At the single-base extension (SBE) site* (Type I) — perturbs extension and dye chemistry, biasing intensity ratios;
  - *Within the probe body* — can reduce binding affinity or alter hybridisation kinetics, especially for common variants.

Table A.1: Technical categories covered by each filtering resource.

egory	**Naeem** A.0.1	**Chen** A.0.2	**Pidsley** A.0.3	**Zhou** A.0.4	**McC**
s-hybridisation / multi-mapping	Yes (Steps 1–2)	Yes	Yes	`MASK_mapping`	T
at CpG / extension base	Yes (Step 5)	–	Yes	`MASK_snp5, MASK_extBase`	
by SNP tolerated	Conditional (Step 6)	–	–	–	
EL / structural variant	Yes (Step 3)	–	–	–	
CpG probes	–	–	Yes	`MASK_nonCG`	

A.0.4 Comprehensive characterization, annotation and innovative use of infinium dna methylation beadchip probes

Zhou *et al.* [40] released a unified probe annotation for both 450K and EPIC arrays with multiple binary mask columns (`MASK_*`). Each mask flags probes to be removed due to specific technical artifacts.

- `MASK.mapping`: probes with low or inconsistent mapping quality (non-unique alignment or presence of INDELs);

- `MASK.typeINextBaseSwitch`: Type-I probes carrying a SNP in the extension base that causes a color-channel switch (*CCS* probes);

- `MASK.extBase`: probes whose extension base is inconsistent with the expected color channel or CpG context;

- `MASK.sub30.copy`: probes with non-unique 30-bp 3' subsequences (potential cross-hybridisation);

- `MASK.snp5.common`: probes overlapping a SNP within ± 5 bp of the interrogated CpG (even with global MAF $< 1\%$);

- `MASK.snp5.GMAF1p`: probes overlapping SNPs with global MAF $> 1\%$;

- `MASK.general`: recommended composite mask integrating mapping, SNP, and cross-reactivity filters for general use;

- `MASK.rmsk15`: probes overlapping RepeatMasker regions (not recommended for exclusion in standard workflows).

This resource provides a programmatic and reproducible way to perform fine-grained probe masking.

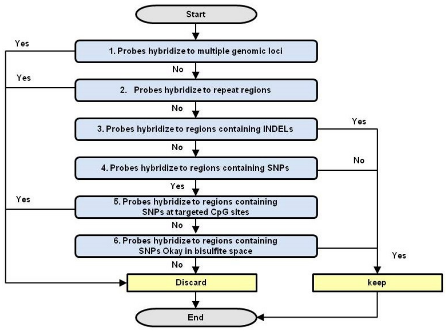


Figure A.1: Workflow for determining affected Probes.

A.0.5 Identification of polymorphic and off-target probe binding sites on the Illumina Infinium MethylationEPIC BeadChip

McCartney *et al.* [42] assessed probe design artifacts and published supplementary tables that include:

- Cross-hybridising CpG-targeting probes;

- Cross-hybridising non-CpG-targeting probes.

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